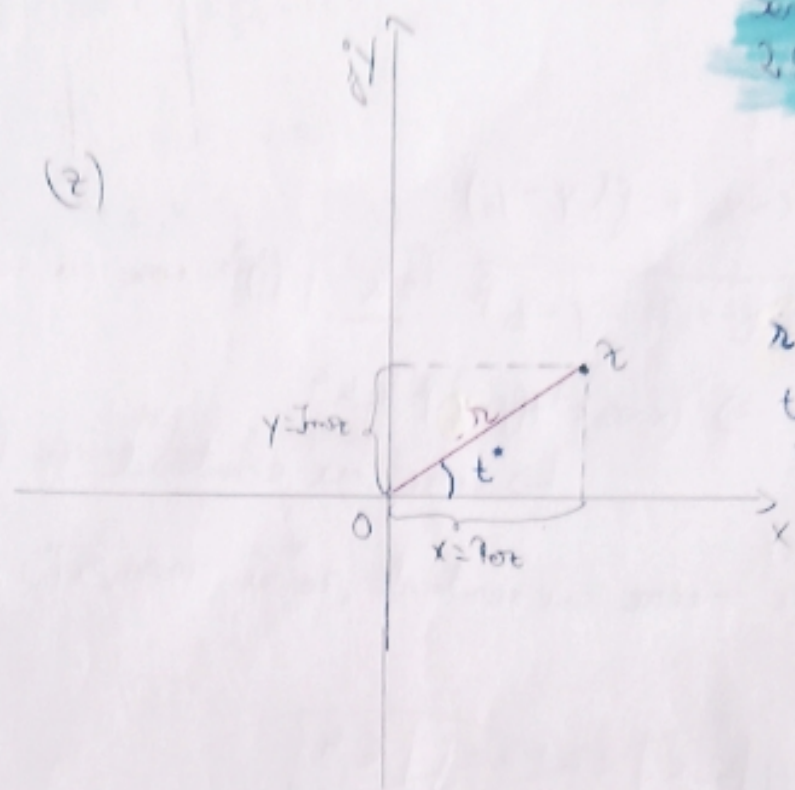


(2)



$$r = |z|$$

$$t^* = \varphi = \theta$$

$$r = \rho = \sqrt{x^2 + y^2} > 0$$

$$x, y \in \mathbb{R}$$

$z = x + jy \rightarrow$ coeficientul părții imaginare
 $y = \text{Im } z$

↓
partea reală

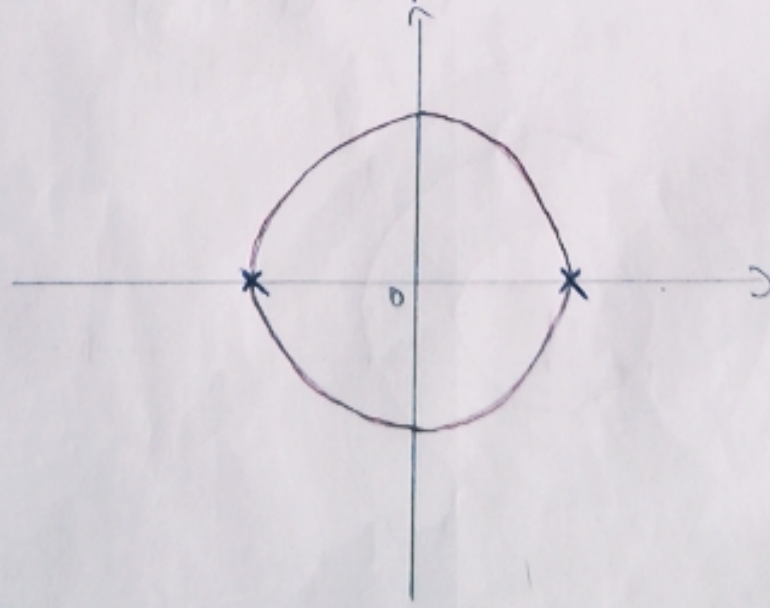
$$x = \text{Re } z$$

$$|z| = \text{const.}$$

$$\sqrt{x^2 + y^2} = k \quad |z|^2$$

$x^2 + y^2 = k^2$ - cerc cu centrul în origine și de rază k.

- doar parte reală când $y = 0$



- în real

- în complex

$$|z - z_0| = k$$

$$|z - z_0| = 0 \Leftrightarrow z - z_0 = 0 \Leftrightarrow z = z_0$$

$$z = x + jy$$

$$z_0 = x_0 + jy_0$$

$$z - z_0 = x - x_0 + j(y - y_0)$$

$$|z - z_0| = \sqrt{(x - x_0)^2 + (y - y_0)^2} = k \quad |()|^2 - \text{cerc cu centrul } z_0 \text{ și rază } k.$$

$$\Rightarrow (x - x_0)^2 + (y - y_0)^2 = k^2$$

$\hookrightarrow$ cerc cu centrul în (x_0, y_0) și de rază k

$$|z - z_0| = r - \text{cerc cu centrul } z_0 \text{ și rază } r.$$

Probleme

Ex 1/25
1. $A_1 = \{z \in \mathbb{C} \mid 2 < |2z + 2 - j| \leq 4\}$

$$|z - z_0| < r$$

$\hookrightarrow$ disc deschis

$$2 < |2(z + 1 - \frac{1}{2}j)| \leq 4 \quad | : 2$$

$$1 < |z + 1 - \frac{1}{2}j| \leq 2$$

$$1 < |z + 1 - \frac{1}{2}j|$$

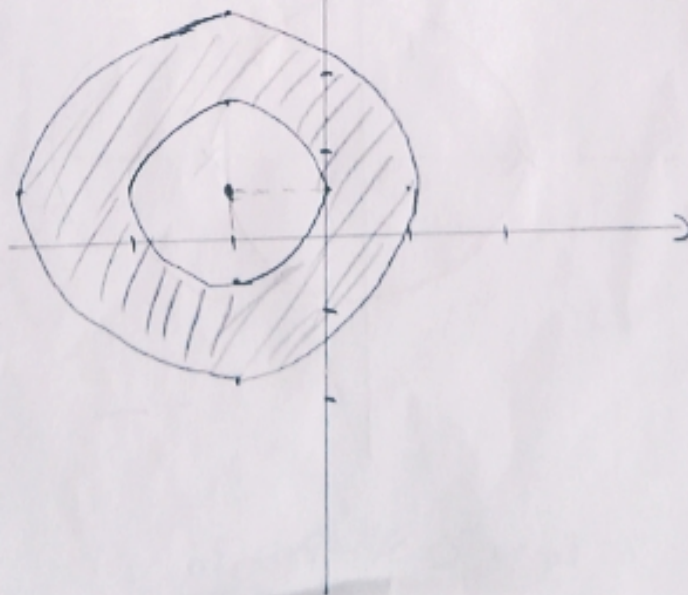
$$C(-1, \frac{1}{2})$$

$$r=1$$

$$|z + 1 - \frac{1}{2}j| \leq 2$$

$$C(-1, \frac{1}{2})$$

$$r=2$$



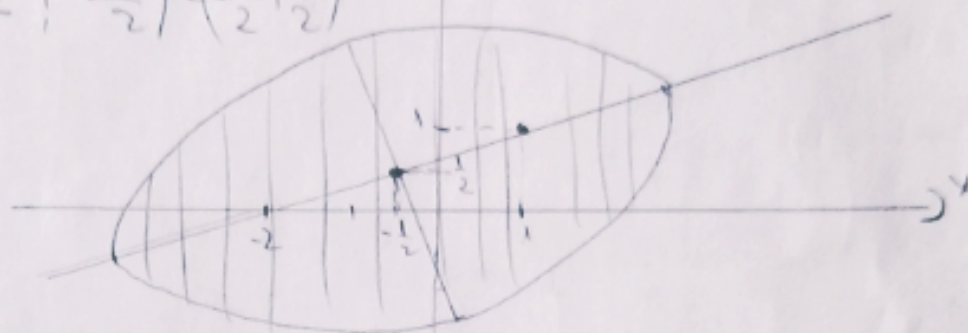
$$2. A_2 = \{(x, y) \mid |z+2| + |z-1-j| = 4\}$$

$$|z+2| + |z-1-j| = 4 \rightarrow 2a$$

dist. de la dkt. $\rightarrow$ dist. de la H(2)
 și $F_2(-2)$ și $F_1(1+j)$

$MF_2 + MF_1 = 2a$ - elipsă

$$M_0\left(-\frac{2+1}{2}; \frac{0+1}{2}\right) = \left(-\frac{1}{2}; \frac{1}{2}\right)$$



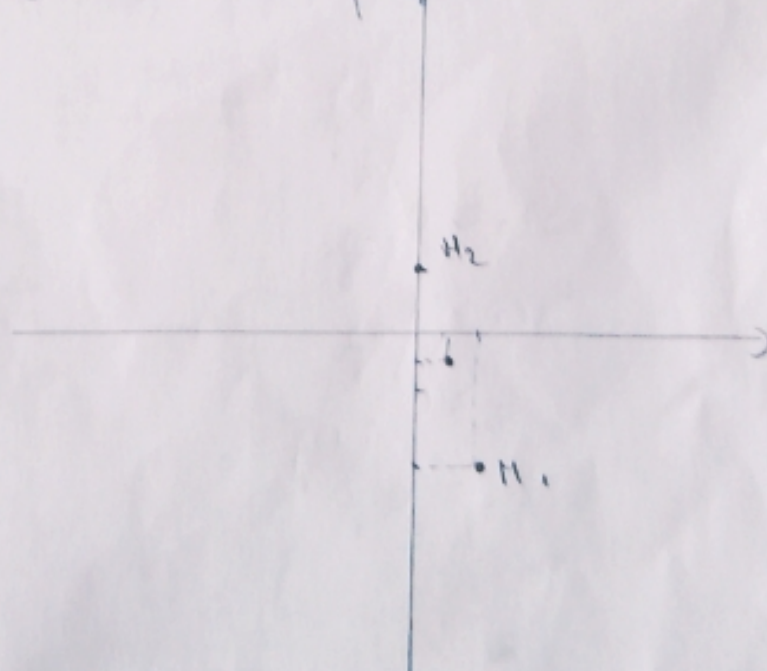
$$3. A_3 = \{(x, y) \mid |z-1+2j| - |z-j| = 3\}$$

$\hookrightarrow$ ramina unei hiperbole

$$M_1 = (1; -2)$$

$$M_2 = (0; 1)$$

$$M\left(\frac{1}{2}; -\frac{1}{2}\right)$$



4. $A_4 = \{ (x, y) \mid |z - 2 + 3j| = |z + 3 - j| \}$ - mediatorea segmentului $M_1 M_2$

$M_1(2; 3)$ $M_2(-3; 1)$

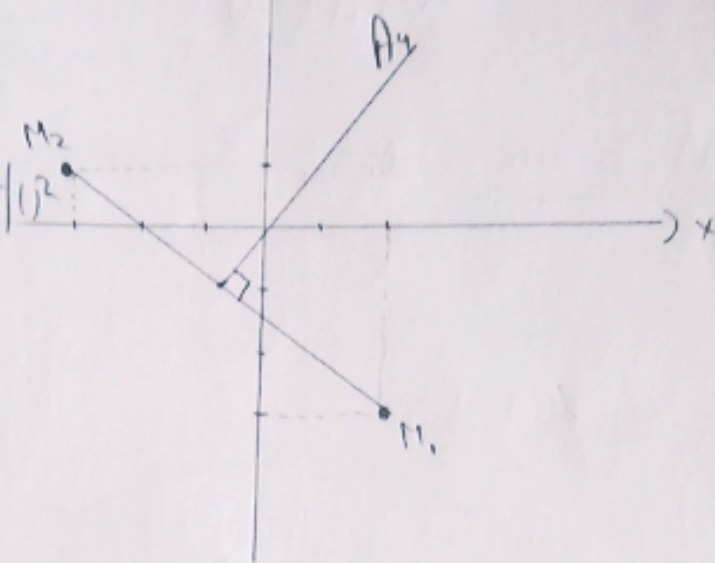
$|z - 2 + 3j| = |z + 3 - j|$

$|x + jy - 2 + 3j| = |x + jy + 3 - j|$

$\sqrt{(x-2)^2 + (y+3)^2} = \sqrt{(x+3)^2 + (y-1)^2} \quad |^2$

~~$x^2 - 4x + 4 + y^2 + 6y + 9 =$~~
 ~~$x^2 + 6x + 9 + y^2 - 2y + 1$~~

$2x - 8y - 3 = 0$



5. Să se determine

a) $\sqrt[4]{j}$

b) $\sqrt[3]{-j}$

c) $\sqrt[5]{1-j}$

d) $\sqrt[9]{\frac{5j-1}{2+3j}}$

a) $j = 1 \left(\cos \frac{\pi}{2} + j \sin \frac{\pi}{2} \right) = 1 \cdot e^{j \frac{\pi}{2}}$
 $1 = \cos 2k\pi + j \sin 2k\pi \quad \forall k \in \mathbb{Z}$

$j = e^{j 2k\pi} \cdot e^{j \frac{\pi}{2}} = e^{j(2k\pi + \frac{\pi}{2})}$

$\sqrt[4]{j} = \sqrt[4]{e^{j(2k\pi + \frac{\pi}{2})}} = e^{\frac{j(2k\pi + \frac{\pi}{2})}{4}}$

$\sqrt[4]{j} = z_k$

$\Rightarrow z_0 = e^{j \frac{\pi}{8}}$

$z_1 = e^{j \frac{5\pi}{8}}$

$z_2 = e^{j \frac{9\pi}{8}}$

$z_3 = e^{j \frac{13\pi}{8}}$

$\sqrt[4]{j} = e^{\frac{j(24\pi + \frac{\pi}{2})}{2}}$
 $e^{j \frac{\pi}{4}}$

$$b) \sqrt[3]{-j} = ?$$

$$-j = 1 \left(\cos \frac{3\pi}{2} + j \sin \frac{3\pi}{2} \right) = 1 \cdot e^{j \frac{3\pi}{2}}$$

! für argumente
regele

$$1 = \cos 2k\pi + j \sin 2k\pi \quad \forall k \in \mathbb{Z}$$

$$\Rightarrow -j = e^{j 2k\pi} \cdot e^{j \frac{3\pi}{2}} = e^{j \pi (2k + \frac{3}{2})}$$

$$\sqrt[3]{-j} = \sqrt[3]{e^{j \pi (2k + \frac{3}{2})}} = e^{j \frac{\pi (2k + \frac{3}{2})}{3}}$$

$$\sqrt[3]{j} = 2k$$

$$\Rightarrow z_0 = e^{j \frac{\pi}{2}} = e^{j \frac{\pi}{2}} = \cos \frac{\pi}{2} + j \sin \frac{\pi}{2} = j$$

$$z_1 = e^{j \frac{7\pi}{6}} = e^{j \frac{7\pi}{6}} = \cos \frac{7\pi}{6} + j \sin \frac{7\pi}{6} = -\cos \frac{\pi}{6} + j \sin \frac{\pi}{6} = -\frac{\sqrt{3}}{2} + j \frac{1}{2}$$

$$z_2 = e^{j \frac{11\pi}{6}} = e^{j \frac{11\pi}{6}} = \cos \frac{11\pi}{6} + j \sin \frac{11\pi}{6} = \cos \left(2\pi - \frac{\pi}{6} \right) + j \sin \left(2\pi - \frac{\pi}{6} \right) = \cos \frac{\pi}{6} - j \sin \frac{\pi}{6} = \frac{\sqrt{3}}{2} - j \frac{1}{2}$$

$$c) \sqrt[5]{1-j} = ?$$

$$1-j = \sqrt{2} \left(\cos \frac{7\pi}{4} + j \sin \frac{7\pi}{4} \right) = \sqrt{2} e^{j \pi (2k + \frac{7}{4})}$$

$$\sqrt[5]{1-j} = \sqrt[5]{\sqrt{2}} e^{j \frac{\pi (2k + \frac{7}{4})}{5}}$$

$$= 2^{\frac{1}{10}} \cdot e^{j \frac{\pi (2k + \frac{7}{4})}{5}}$$

$$\forall k \in \{0, 1, 2, 3, 4\}$$

$$d) \sqrt[n]{\frac{5j-1}{2+3j}} \quad ?$$

$$\sqrt[n]{\frac{(2-3j)(5j-1)}{4+9}} = \sqrt[n]{\frac{10j-2+15+3j}{13}} = \sqrt[n]{\frac{13+13j}{13}} = \sqrt[n]{1+j}$$

$$1+j = \sqrt{2} \left(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4} \right) = \sqrt{2} e^{j\frac{\pi}{4}}$$

$$\sqrt[n]{1+j} = \sqrt[n]{\sqrt{2}} e^{j\frac{\pi}{n} \left(2k + \frac{1}{4} \right)} = 2^{\frac{1}{2n}} e^{j\frac{\pi}{n} \left(2k + \frac{1}{4} \right)} \quad \forall k \in \overline{0; n-1}$$

Reprezentare

- vârfurile poligonului cu 10 laturi, inscrie în cerc.

6. Să se rezolve ecuația — ecuație de gradul $(n-1)$

$$\frac{(z+1+j)^n}{(z+1-j)^n} = 1 \quad | : (z+1-j)^n$$

$$\frac{(z+1+j)^n}{(z+1-j)^n} = 1$$

$$\left(\frac{z+1+j}{z+1-j} \right)^n = 1$$

$$\sum = 1 = e^{2k\pi j} \quad \forall k \in \mathbb{Z}$$

$$z_k = e^{\frac{2k\pi j}{n}} \quad \forall k \in \overline{0; n-1}$$

$$\widehat{\parallel} \quad k \in \overline{1; n-1}$$

$$(z+1+j)^n = (z+1-j)^n$$

$$\cancel{z^n} + \binom{n}{1} z^{n-1} (1+j) + \binom{n}{2} z^{n-2} (1+j)^2 + \dots = \cancel{z^n} + \binom{n}{1} z^{n-1} (1-j) + \dots$$

$$z_0 = e^0 = 1 \Rightarrow \frac{z+1+j}{z+1-j} = 1 \quad (\Rightarrow) \quad z+1+j = z+1-j \quad \parallel -z-1$$

$$\frac{z_k + 1 + j}{z_k + 1 - j} = e^{\frac{2k\pi j}{n}}; k \in \overline{1, n-1}$$

$$z_k = ?$$

$$z_k + 1 + j = e^{\frac{2k\pi j}{n}} (z_k + 1 - j)$$

$$z_k + \sqrt{2} e^{j\pi(2k+1/4)} = z_k e^{\frac{2k\pi j}{n}} + e^{\frac{2k\pi j}{n}} \cdot \sqrt{2} \cdot e^{j\pi(2k+7/4)}$$

$$z_k \left(1 - e^{\frac{2k\pi j}{n}} \right) = \sqrt{2} \left(e^{\frac{2k\pi j}{n} + j\pi(2k+7/4)} - e^{j\pi(2k+1/4)} \right) / e^{\frac{2k\pi j}{n}}$$

$$z_k \left(1 - e^{\frac{2k\pi j}{n}} \right) = \sqrt{2} \left(e^{\frac{k}{n} + k + \frac{7}{8}} - e^{k + \frac{1}{8}} \right)$$

$$z_k = \frac{\sqrt{2} (e^{\frac{k}{n} + \frac{7}{8}} - e^{\frac{1}{8}})}{1 - e^{\frac{2k\pi j}{n}}}$$

Temă.

$$(2-3j)z^6 + 1 + 5j = 0 \quad \sqrt[6]{}$$

$$z^8 + (2j-1)z^4 - j - 1 = 0 \quad z^4 = u$$

$$z^6 = \frac{-1-5j}{2-3j} \quad \sqrt[6]{}$$

$$z = \sqrt[6]{\frac{(2+3j)(-1-5j)}{4+9}} = \sqrt[6]{\frac{-2-10j-3j^2+15}{13}} = \sqrt[6]{\frac{13-13j}{13}} = \sqrt[6]{1-j}$$

$$1-j = \sqrt{2} e^{j\pi(2k+7/4)}$$

$$\Rightarrow \sqrt[6]{1-j} = 2^{1/12} e^{j\pi(2k+7/4)/6}$$

$$k \in \overline{0, 5}$$

$$\begin{aligned} z_0 &= 2^{1/12} e^{j\frac{7\pi}{24}} \\ z_1 &= 2^{1/12} e^{j\frac{13\pi}{24}} \\ z_2 &= 2^{1/12} e^{j\frac{19\pi}{24}} \\ z_3 &= 2^{1/12} e^{j\frac{25\pi}{24}} \\ z_4 &= 2^{1/12} e^{j\frac{31\pi}{24}} \\ z_5 &= 2^{1/12} e^{j\frac{37\pi}{24}} \end{aligned}$$

$$z^8 + (2j-1)z^4 - j - 1 = 0$$

$$z^4 = t \Rightarrow t^2 + (2j-1)t - j - 1 = 0$$

$$\Delta = (2j-1)^2 - 4 \cdot 1 \cdot (-j-1) = -4 - 4j + 1 + 4j + 4 = 1$$

$$\Rightarrow t_1 = \frac{1-2j+j}{2} = \frac{1}{2} - \frac{1}{2}j \quad r = \frac{\sqrt{2}}{2}$$

$$t_2 = \frac{1-2j-j}{2} = \frac{1}{2} - \frac{3}{2}j \quad r_2 = \frac{\sqrt{10}}{2}$$

$$z_1 = \sqrt[4]{\frac{1}{2} - \frac{1}{2}j}$$

$$z_1 = \sqrt[4]{\frac{\sqrt{2}}{2}} e^{j\pi(2k+\frac{1}{4})} = 2^{-\frac{1}{4}} \cdot e^{j\pi(2k+\frac{1}{4})} \quad \forall k \in \overline{0,3}$$

$$z_2 = \sqrt[4]{\frac{1}{2} - \frac{3}{2}j}$$

$$z_2 = \sqrt[4]{\frac{\sqrt{10}}{2}} e^{j\pi(2k+\frac{5}{3})}$$

$$z_2 = 10^{\frac{1}{8}} \cdot 2^{-\frac{1}{4}} \cdot e^{j\pi(2k+\frac{5}{3})}$$

$$z_2 = 2^{-\frac{1}{8}} \cdot 5^{\frac{1}{8}} \cdot e^{j\pi(2k+\frac{5}{3})}$$

$$= 2^{-\frac{1}{8}} \cdot 5^{\frac{1}{8}} \cdot e^{j\pi(2k+\frac{5}{3})}$$

1. Să se rezolve ecuația:
 $z^n = \bar{z}$ ($n \in \mathbb{N}$)

$$n=0 \Rightarrow 1 = \bar{z} \Rightarrow z=1$$

$$n=1 \Rightarrow z = \bar{z} \Rightarrow z \in \mathbb{R}$$

$$z^n = \bar{z}; n \in \{2, 3, \dots\}$$

$$|z^n| = |\bar{z}|$$

$$|z|^n = |z|$$

$$|z|(|z|^{n-1} - 1) = 0$$

$$|z| = 0 \Rightarrow z = 0$$

$$\text{sau } |z|^{n-1} = 1$$

$$|z| = 1 \Rightarrow z = \cos \varphi + j \sin \varphi = e^{j\varphi}$$

obs

$$x^{n-1} - 1 = 0$$

$$x = 1 \text{ dacă } n-1 \text{ impar}$$

$$x = -1 \text{ dacă } n-1 \text{ par}$$

$$x^{2k+1} - 1 = 0$$

$$x^{2k} - 1 = (x^2 - 1)(x^{2k-2} + x^{2k-4} + \dots + 1) = 0$$

$$(x-1)(x^{2k} + x^{2k-2} + \dots + 1) = 0$$

rad. complexe

$$\Rightarrow e^{j\theta} = e^{-j\theta} \quad | \quad e^{j\theta}$$

$$e^{j(n+1)\theta} = 1 = e^{j2k\pi}$$

$$z = e^{j\theta} = e^{\frac{2k\pi j}{n+1}}$$

$$k = 0, 1, \dots, n$$

- $n+1$ rădăcini

dar $z=0$ - rădăcină

$\frac{n+2}{2}$ rădăcini
în total

2. Să se calculeze:

$$a) \operatorname{Ln} \frac{1-j}{\sqrt{2}} =$$

$$\operatorname{Ln} z = \ln r + j\theta; z \neq 0$$

$\hookrightarrow$ determinarea principală

$$\operatorname{Ln} z = \ln r + j(\theta + 2k\pi)$$

$$= \ln |z| + j(\arg z + 2k\pi)$$

$$\rho = 1$$

$$\theta = \frac{7\pi}{4}$$

$$\operatorname{Ln} \frac{1-j}{\sqrt{2}} = \ln 1 + j\left(\frac{7\pi}{4} + 2k\pi\right) = j\left(\frac{7\pi}{4} + 2k\pi\right) ; k \in \mathbb{Z}$$

$$b) (-1)^j = e^{-1 \operatorname{Ln} j} = e^{j \operatorname{Ln}(-1)}$$

$$\boxed{\begin{aligned} z^a &= e^{z \operatorname{Ln} a} \\ z^a &= e^{a \operatorname{Ln} z} \end{aligned}}$$

$$\operatorname{Ln}(-1) = \ln|-1| + j(\pi + 2k\pi) = j\pi(2k+1) ; k \in \mathbb{Z}$$

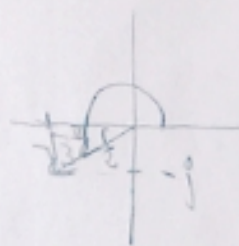
$\Downarrow$
 $\in \text{OY}$

3. Să se precizeze partea reală și imaginară pt del. principale pentru:

$$a) \operatorname{Ln}(-\sqrt{3}-j) = \ln 2 + j \frac{7\pi}{6}$$

$$\Rightarrow \operatorname{Re} = \ln 2$$

$$\operatorname{Im} = \frac{7\pi}{6}$$



! Atenție.

$$4. z = (-1)^{j+1}$$

$$\operatorname{Ln}(-1) = \ln|-1| + j\pi = j\pi$$

$$= e^{(j+1) \operatorname{Ln}(-1)}$$

$$= e^{(j+1) \cdot j\pi} = e^{-\pi(-1+j)} = e^{-\pi} (\cos \pi + j \sin \pi)$$

$$= e^{-\pi} (-1 + 0) = -e^{-\pi} = \operatorname{Re} z$$

$$\Rightarrow \operatorname{Im} z = 0.$$

Funcții armonice

$$f(z) = P(x,y) + iQ(x,y)$$

C-R $\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases} \quad M_0(x_0, y_0) \Rightarrow f \text{ - meromorfă în } M_0$

$f(z) \in D \subseteq \mathbb{C} \Rightarrow f \text{ - armonică pe } D.$

i. $f(z) = z^3 + 2z^3 + z^2 - 3\bar{z}^2 + 20z + 15\bar{z} + 1 + i$

- puncte de meromorfitate

$\Rightarrow z = x + iy$

$$\begin{aligned} f(z) &= (x - iy)^3 + 2(x + iy)^3 + (x + iy)^2 - 3(x - iy)^3 + 20(x + iy) + 15(x - iy) + 1 + i \\ &= \underline{x^3 - 3ix^2y - 3xy^2 + iy^3} + 2(\underline{x^3 + 3ix^2y - 3xy^2 - iy^3}) + \\ &\quad x^2 + 2xyi - y^2 - 3(x^2 - 2xyi - y^2) + 20x + 20iy + 15x - 15iy + 1 + i \\ &= 3x^3 + 3ix^2y - 3xy^2 - iy^3 - 2x^2 + 8xyi + 2y^2 + 35x + 5iy + 1 + i \\ &= 3x^3 - 3xy^2 - 2x^2 + 2y^2 + 35x + 1 + i(3x^2y - y^3 + 8xy + 5y + 1) \end{aligned}$$

$\Rightarrow \operatorname{Re} f(z) = 3x^3 - 3xy^2 - 2x^2 + 2y^2 + 35x + 1$

$\operatorname{Im} f(z) = 3x^2y - y^3 + 8xy + 5y + 1$

$\frac{\partial P}{\partial x} = 9x^2 - 6y^2 - 4x + 35$

$\Rightarrow 6x^2 - 6y^2 - 12x + 80 = 0 \quad | :6$

$\frac{\partial Q}{\partial y} = 3x^2 - 3y^2 + 8x + 5$

$x^2 - y^2 - 2x + 5 = 0$

$\frac{\partial P}{\partial y} = -12xy + 4y$

$-12xy + 12y = 0 \quad | :12$

$\frac{\partial Q}{\partial x} = 6xy + 8y$

$xy - y = 0$

$$\Rightarrow \begin{cases} y(x-1)=0 \\ x^2-y^2-2x+5=0 \end{cases}$$

$$y=0 \quad \text{ou} \quad x-1=0 \quad \Rightarrow \quad 1-y^2-2+5=0$$

$\Downarrow$

$$x^2-2x+5=0$$

$$\Delta = 4-4 \cdot 5 = -16 < 0$$

nenhuma sol. real

$$y^2 = 4$$

$$y = \pm 2$$

$$\Rightarrow S = \{(1, -2); (1, +2)\}$$

$$\Rightarrow \boxed{\begin{matrix} z_1 = 1+2j \\ z_2 = 1-2j \end{matrix}} \text{ - pontos de monogenidade}$$

Obs!

$$f(z) = F(z, \bar{z})$$

$$\frac{\partial F}{\partial z} = F'(z)$$

$$\frac{\partial F}{\partial \bar{z}} = 0$$

O função monogênica ou derivada falha de $\bar{z}$, nula.

$$b) f(z) = F(z, \bar{z}) = |z|^{10} - \frac{1}{2} \bar{z} \\ = z^5 \cdot \bar{z}^5 - \frac{1}{2} \bar{z}$$

$$\frac{\partial F}{\partial \bar{z}} = 5z^5 \cdot \bar{z}^4 - \frac{1}{2} = 0$$

$$\boxed{|z|^2 = z \cdot \bar{z}} \Leftarrow$$

$$z = x + jy$$

$$10z^5 \cdot \bar{z}^4 = 1 \cdot \bar{z}$$

$$\Rightarrow \underbrace{5|z|^{10} - \frac{1}{2}x + \frac{1}{2}jy}_{\in \mathbb{R}} = 0$$

$$\begin{cases} 5|z|^{10} - \frac{1}{2}x = 0 \\ \frac{1}{2}y = 0 \Rightarrow |z|^{10} = |x|^{10} \\ 5|z|^{10} - \frac{1}{2}y = 0 \end{cases}$$

$$x < 0 \Rightarrow x = -t$$

$$5t^{10} + \frac{1}{2}t = 0$$

$$t = 0 \quad \text{sau} \quad t^9 + \frac{1}{10} = 10 \quad t = -\sqrt[9]{\frac{1}{10}}$$

$$x > 0$$

$$5t^{10} - \frac{1}{2}x = 0$$

$$x = 0 \quad \text{sau} \quad x = \sqrt[9]{\frac{1}{10}}$$

$$\Rightarrow z_0 = 0$$

$$z_{1,2} = \pm \sqrt[9]{\frac{1}{10}}$$

$$\frac{\partial f}{\partial z} = 5z^4 \cdot \bar{z} \quad \Bigg|_{z_0, z_1, z_2}$$

$$f'(z_0) = 0$$

$$f(z_{1,2}) = 5x_{1,2}^9 = 5\left(\pm \frac{1}{10}\right) = \pm \frac{1}{2}$$

2° Sa se det. a, b, c, d a.t. funcția să fie olomorfa pe

$$f(z) = \underbrace{ax^2 + bxy + cy^2 + x - y}_{\mathcal{P}} + i \underbrace{(ax^2 + 2xy + dy^2 + x + ay)}_{\mathcal{Q}} \quad x, y \in \mathbb{R}^2$$

$$\frac{\partial \mathcal{P}}{\partial x} = 2ax + by + 1$$

$$\Rightarrow 2x(1-a) + y(2d-b) + a - 1 = 0$$

$$\frac{\partial \mathcal{Q}}{\partial y} = 2x + 2dy + a$$

$$\frac{\partial \mathcal{P}}{\partial y} = bx + 2cy - 1$$

$$\Rightarrow x(2a-b) + y(2-2c) + 2 = 0$$

$$\frac{\partial \mathcal{Q}}{\partial x} = 2ax + 2y + 1$$

$$\begin{cases} z - za = za - b \\ za - b = z - za \\ a - 1 = 2 \end{cases} \Rightarrow \begin{cases} a = 1 \\ b = -2 \\ c = -1 \\ d = -1 \end{cases}$$

$$f(z) = \underline{x^2} - 2xy - \underline{y^2} + \underline{1} - y + j(\underline{x^2} + 2xy - \underline{y^2} + \underline{1} + y)$$

$$\begin{aligned} z &= x + jy \\ \bar{z} &= x - jy \end{aligned} \Rightarrow \boxed{\begin{cases} x = \frac{z + \bar{z}}{2} \\ y = \frac{z - \bar{z}}{j} \end{cases}} \quad !$$

$$\begin{aligned} f(z) &= (x + jy)^2 + j(x + jy)^2 + (x + jy) + j(x + jy) \\ &= z^2 + jz^2 + z + jz \\ &= z^2(1+j) + z(1+j) \\ &= (1+j)(z^2 + z) \end{aligned}$$

3. Să se determine funcția derivată $f(z) = P + jQ$ știind că $P(x, y) = \frac{x}{x^2 + y^2} - 2xy$

$$\Delta P = 0 \quad \Rightarrow \quad \Delta Q = 0$$

C-R

$$dQ = \frac{\partial Q}{\partial x} dx + \frac{\partial Q}{\partial y} dy = -\frac{\partial P}{\partial y} dx + \frac{\partial P}{\partial x} dy$$

$$\downarrow$$

$$\frac{\partial}{\partial y} = -\frac{\partial}{\partial x}$$

def. lui Cauchy

$$-\frac{\partial^2 P}{\partial y^2} = \frac{\partial^2 P}{\partial x^2} \quad (\Rightarrow \Delta P = 0 \Rightarrow)$$

$$Q(x, y) = - \int_{x_0}^x \left(\frac{\partial P}{\partial y} \right) (t, y_0) dt + \int_{y_0}^y \left(\frac{\partial P}{\partial x} \right) (x_0, t) dt + C$$

$$\text{Für } x_0 = 0 \quad \Rightarrow$$

$$y_0 = 1$$

p. Sa se det. functia olomomorfa

$$f(z) = P(x,y) + jQ(x,y)$$

$$Q(x,y) = e^x (x \sin y + y \cos y); f(1) = e$$

$$\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = e^x (x \cos y + \cos y - y \sin y)$$

$$\frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = -e^x (x \sin y + y \cos y) - e^x \sin y$$

$$dQ = \frac{\partial Q}{\partial x} dx + \frac{\partial Q}{\partial y} dy$$

$$= \underbrace{e^x (x \sin y + y \cos y + \sin y)}_0 dx + \underbrace{e^x (x \cos y + \cos y - y \sin y)}_V dy$$

$$\frac{\partial V}{\partial y} = \frac{\partial V}{\partial x} \Leftrightarrow \Delta P = 0$$

$$\begin{aligned} Q(x,y) &= \int_{x_0}^x e^x (x \sin y + y \cos y + \sin y) dx + \int_{y_0}^y e^x (x \cos y + \cos y - y \sin y) dy \\ &= \int_{y_0}^y e^x (x \cos y + \cos y - y \sin y) dy + \underbrace{\int_{x_0}^x e^x (x \sin y_0 + y \cos y_0 + \sin y_0) dx}_{=0} \end{aligned}$$

$$(x_0, y_0) \in D \subseteq \mathbb{R}^2$$

$$y_0 = 0, x_0 = 0$$

$$\Rightarrow Q(x,y) = x e^x \int_0^y \cos y dy + e^x \underbrace{\int_0^y (\cos y - y \sin y) dy}_{y \cos y}$$

$$= x e^x [\sin y]_0^y + e^x [y \cos y]_0^y$$

$$= x e^x \sin y + e^x y \cos y + C$$

$$P(x,y) = - \int_{y_0}^y e^x (x \sin y + y \cos y + \sin y) dy + \int_{x_0}^x [e^x (x \cos y + \cos y - y \sin y)] dx$$

$$(x_0, y_0) = (0, 0)$$

$$\Rightarrow P(x,y) = -x e^x \int_0^y \sin y dy - e^x \int_0^y (y \cos y + \sin y) dy + \int_0^x e^x (x+1) dx$$

$$= x e^x [\cos y]_0^y - e^x y \sin y + x e^x + C$$

$$= x e^x (\cos y - 1) - e^x y \sin y + x e^x + C$$

$$= x e^x \cos y - e^x y \sin y + C$$

$$f(z) = P(x,y) + Q f(x,y)$$

$$= e^x [x \cos y - y \sin y + \int (x \sin y + y \cos y)] + C$$

$$= e^x [\sin y (x - y) + \cos y (x + y)] + C$$

$$= e^x [y \sin y (x + y) + \cos y (x + y)] + C$$

$$= e^x (x + y) (\cos y + y \sin y) + C$$

$$= z \cdot e^x \cdot e^{iy}$$

$$= z \cdot e^z + C$$

$$f(z) = z e^z + C$$

$$f(1) = e \Rightarrow 1 \cdot e + C = e \Rightarrow C = 0$$

$$\Rightarrow f(z) = z \cdot e^z$$

$$2. \quad P - Q = \frac{x-y}{2} \ln(x^2+y^2) - (x+y) \operatorname{arctg} \frac{y}{x} ; \quad p(x,y)$$

$$\begin{cases} \frac{\partial P}{\partial x} - \frac{\partial Q}{\partial x} = \frac{\partial \varphi}{\partial x} \\ \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial y} = \frac{\partial \varphi}{\partial y} \end{cases}$$

$$\begin{cases} \frac{\partial P}{\partial x} + \frac{\partial P}{\partial y} = \frac{\partial \varphi}{\partial x} \\ \frac{\partial P}{\partial y} - \frac{\partial P}{\partial x} = \frac{\partial \varphi}{\partial y} \end{cases}$$

$$\frac{\partial P}{\partial y} = \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} \right)$$

$$\frac{\partial P}{\partial x} = \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} - \frac{\partial \varphi}{\partial y} \right)$$

$$dP = \frac{\partial P}{\partial x} dx + \frac{\partial P}{\partial y} dy$$

$$= \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} - \frac{\partial \varphi}{\partial y} \right) dx + \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} + \frac{\partial \varphi}{\partial y} \right) dy$$

$$\frac{\partial \varphi}{\partial x} = \frac{1}{2} \ln(x^2+y^2) + \frac{(x-y)x}{x^2+y^2} - \operatorname{arctg} \frac{y}{x} + \frac{(x+y)y}{x^2+y^2} \quad (+)$$

$$\frac{\partial \varphi}{\partial y} = -\frac{1}{2} \ln(x^2+y^2) + \frac{(x-y)y}{x^2+y^2} - \operatorname{arctg} \frac{y}{x} - \frac{(x+y)x}{x^2+y^2} \quad (-)$$

$$\frac{1}{2} \ln(x^2+y^2) - \operatorname{arctg} \frac{y}{x} + 1 + \frac{1}{2} \ln(x^2+y^2) + \operatorname{arctg} \frac{y}{x} + 1$$

$$\Rightarrow \varphi(x,y) = \frac{1}{2} \varphi(x,y) + \int_{x_0}^x \left[\frac{1}{2} \ln(x^2+y^2) + 1 + \operatorname{arctg} \frac{y}{x} \right] dx +$$

$$+ \int_{y_0}^y \left[\frac{1}{2} \ln(x_0^2+y^2) + 1 - \operatorname{arctg} \frac{y}{x_0} \right] dy$$

$$3. f(z) = P(x,y) + j Q(x,y) \quad - \text{olamorfia}$$

$$P(x,y) = \varphi(x^2+y^2)$$

$$\varphi \in C^2(\mathbb{R})$$

$$\Rightarrow \begin{cases} \Delta P = 0 \\ \Delta Q = 0 \end{cases}$$

$$\Delta P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2}$$

$$\frac{\partial P}{\partial x} = \varphi'(x^2+y^2) \cdot 2x$$

$$\frac{\partial P}{\partial y} = \varphi'(x^2+y^2) \cdot 2y$$

$$\frac{\partial^2 P}{\partial x^2} = 2\varphi'(x^2+y^2) + 4x^2\varphi''(x^2+y^2)$$

$$\frac{\partial^2 P}{\partial y^2} = 2\varphi'(x^2+y^2) + 4y^2\varphi''(x^2+y^2)$$

(+)

$$0 = 4\varphi'(x^2+y^2) + 4\varphi''(x^2+y^2) \cdot (x^2+y^2) \quad | : 4$$

$$x^2+y^2 = t \Rightarrow t \cdot \varphi''(t) + \varphi'(t) = 0$$

$$\varphi'(t) = u(t)$$

$$t \cdot u'(t) + u(t) = 0$$

$$t \frac{du}{dt} = -u$$

$$\frac{du}{u} = -\frac{dt}{t}$$

$$\ln|u| = -\ln|t| + \ln|k|$$

$$u(t) = \frac{C_1}{t}$$

$$\varphi'(t) = \frac{C_1}{t} \Rightarrow \boxed{\varphi(t) = C_1 \ln|t| + C_2}$$

$$P(x,y) = C_1 \ln(x^2+y^2) + C_2$$

$$\frac{\partial P}{\partial x} = C_1 \frac{2x}{x^2 y^2} = \frac{\partial Q}{\partial y}$$

$$\frac{\partial P}{\partial y} = C_1 \frac{2y}{x^2 y^2} = \frac{\partial Q}{\partial x}$$

$$dQ(x, y) = -C_1 \frac{2y}{x^2 y^2} dx + C_1 \frac{2x}{x^2 y^2} dy$$

$$Q(x, y) = -2C_1 y \int_0^x \frac{dx}{x^2 y^2} + 2C_1 x \int_0^y \frac{dy}{y^2}$$

$$= -2C_1 x \cdot \frac{1}{x} \left[\operatorname{arctg} \frac{x}{y} \right]_0^x + C_3'$$

$$= -2C_1 \operatorname{arctg} \frac{x}{y} + C_3'$$

$$= -2C_1 \left(\frac{\pi}{2} - \operatorname{arctg} \frac{y}{x} \right) + C_3'$$

$$\operatorname{Ln} z = \operatorname{Im}|z| + j(\operatorname{arg} z + 2k\pi)$$

$$\Rightarrow Q(x, y) = 2C_1 \operatorname{arctg} \frac{y}{x} + \underbrace{C_3' - C_1 \pi}_{= C_3}$$

$$\operatorname{arctg} x + \operatorname{arctg} \frac{1}{x} = a \quad | \operatorname{tg}$$

$$\operatorname{tg}(\operatorname{arctg} x) + \operatorname{tg}(\operatorname{arctg} \frac{1}{x}) = \operatorname{tg} a$$

$$\operatorname{tg}(\operatorname{arctg} x) + \operatorname{tg}(\operatorname{arctg} \frac{1}{x})$$

$$1 - \operatorname{tg}(\operatorname{arctg} x) \cdot \operatorname{tg}(\operatorname{arctg} \frac{1}{x})$$

$$= \operatorname{tg} a$$

$$\Rightarrow a = \frac{\pi}{2}$$

$$\Rightarrow f(z) = C_1 \operatorname{Im}(x^2 + y^2) + C_2 + 2jC_1 \operatorname{arctg} \frac{y}{x} + jC_3$$

$$= 2C_1 \left[\operatorname{Im}|z| + j \operatorname{arctg} \frac{y}{x} \right] + C_2 + jC_3$$

$$f(z) = 2 \operatorname{Arg} z + K$$

$$C, \in \mathbb{R}$$

$$K = C_2 + j C_3 \in \mathbb{C}$$

$\hookrightarrow$ constantă complexă

4. Să se determine: coeficientul de deformare liniară și unghiul de rotație

$$Z = \frac{2z - j + 4}{2 + 2j}; \quad z_0 = j$$

$$Z' = f'(z)$$

$$Z'_0 = f'(z_0) \quad \begin{cases} |f'(z_0)| - \text{coeficientul de deformare} \\ \arg f'(z_0) - \text{unghiul de rotație} \end{cases}$$

$$f'(z_0) \neq 0$$

$$\begin{aligned} Z' &= \frac{2(2+jz) - j(2z-j+4)}{(2+2j)^2} \\ &= \frac{4 + 2jz - 2jz + 1 - 4j}{(2+jz)^2} \\ &= \frac{3-4j}{(2+jz)^2} \end{aligned}$$

$$Z'_0 = \frac{3-4j}{(2-j)^2} = 3-4j$$

$$\begin{aligned} \Rightarrow \text{coeficientul de deformare:} & \sqrt{3^2 + (-4)^2} = 5 \\ \text{unghiul de rotație:} & \arg(-\frac{4}{3}) \end{aligned}$$

(ii) reglurile din plan în care se poate scrie

$$z' = \frac{3-4j}{(2+jz)^2}$$

$$|z'| > 1$$

$$\left| \frac{3-4j}{(2+jz)^2} \right| > 1$$

$$\underline{|3-4j| = |2+jz|^2}$$

$$5 = |2+jz|^2$$

$$|2+jz| = \sqrt{5} \text{ - frontiera}$$

$$|j(z-2j)| = \sqrt{5}$$

$$|z-2j| = \sqrt{5} \Rightarrow |z-2j| < \sqrt{5}$$

$$|z-z_0| = r \Rightarrow \text{arc } r = \sqrt{5} \text{ și centru } (0,2)$$

$$\left| \frac{3-4j}{(2+jz)^2} \right| > 1 \Rightarrow \text{interiorul cercului de} \\ \text{raza } r = \sqrt{5} \text{ și centru } (0,2)$$

1. Să se determine o funcție olomorfa $f(z) = P(x, y) + jQ(x, y)$

$$P(x, y) = \frac{x}{2} \ln(x^2 + y^2) - y \operatorname{arctg} \frac{y}{x}; \quad f(1) = j$$

$$\frac{\partial P}{\partial x} = \frac{1}{2} \ln(x^2 + y^2) + \frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2}$$

$$= \frac{1}{2} \ln(x^2 + y^2) + 1$$

$$\frac{\partial P}{\partial y} = \frac{xy}{x^2 + y^2} - \operatorname{arctg} \frac{y}{x} - \frac{xy}{x^2 + y^2} = -\operatorname{arctg} \frac{y}{x} = \frac{\partial Q}{\partial x}$$

$$f'(z) \stackrel{\text{olomorfa}}{=} \frac{\partial P}{\partial x} + j \frac{\partial Q}{\partial x} = \frac{1}{2} \ln(x^2 + y^2) + 1 + j \operatorname{arctg} \frac{y}{x}$$

$$z = x$$

$$y = 0$$

$$= \left(\ln(\sqrt{x^2 + y^2}) + j \operatorname{arctg} \frac{y}{x} \right)$$

$$= \ln|z| + j \arg z + 1$$

$$= \ln z + 1$$

$$f'(z) = \ln z + 1$$

$$f(z) = \int (\ln z + 1) dz$$

$$f'(x) = \ln x + 1$$

$$f(x) = \int (\ln x + 1) dx$$

$$u = \ln x$$

$$u' = \frac{1}{x}$$

$$g' = 1 \Rightarrow g = x$$

$$\Rightarrow f(x) = x \ln x - x + x + C = x \ln x + C$$

$$\Rightarrow f(z) = z \cdot \ln z + C$$

$$f(1) = 1 \ln 1 + C$$

$$\Rightarrow C = j$$

$$f(z) = z \ln z + j$$

$$f(z) = P(x, y) + iQ(x, y) = u(n, t) + jv(n, t)$$

$$\begin{cases} x = n \cos t \\ y = n \sin t \end{cases}$$

$$C-R \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} \end{cases}$$

$$\frac{\partial P}{\partial x} = \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x} = \frac{\partial u}{\partial n} \cos t + \frac{\partial u}{\partial t} \cdot \frac{\sin t}{n}$$

$$\frac{\partial P}{\partial y} = \frac{\partial u}{\partial n} \cdot \frac{\partial n}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} = \frac{\partial u}{\partial n} \sin t + \frac{\partial u}{\partial t} \cdot \frac{\cos t}{n}$$

$$r = \sqrt{x^2 + y^2}$$

$$t = \arctan \frac{y}{x} + \varepsilon \pi \quad (\varepsilon = 0 \text{ pt cad. I of } \pi)$$

$$\frac{\partial n}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{n \cos t}{n} = \cos t$$

$$\frac{\partial n}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} = \sin t$$

$$\frac{\partial t}{\partial x} = -\frac{y}{x^2} \cdot \frac{x^2}{x^2 + y^2} = -\frac{n \sin t}{n^2} = -\frac{\sin t}{n}$$

$$\frac{\partial t}{\partial y} = \frac{\frac{1}{x}}{1 + \frac{y^2}{x^2}} = \frac{x}{x^2 + y^2} = \frac{n \cos t}{n^2} = \frac{\cos t}{n}$$

$$\begin{cases} \frac{\partial P}{\partial x} = \frac{\partial v}{\partial n} \cdot \sin t + \frac{\partial v}{\partial t} \cdot \frac{\cos t}{n} = \frac{\partial Q}{\partial y} \end{cases}$$

$$\begin{cases} \frac{\partial P}{\partial y} = -\frac{\partial v}{\partial n} \cdot \cos t + \frac{\partial v}{\partial t} \cdot \frac{\sin t}{n} = -\frac{\partial Q}{\partial x} \end{cases}$$

$$\Rightarrow \frac{\partial u}{\partial r} = \frac{\partial v}{\partial t} \cdot \frac{1}{r}$$

$$\Rightarrow \frac{1}{r} \frac{\partial u}{\partial t} = - \frac{\partial v}{\partial r}$$

Exemple:

$$f(z) = z^n \text{ - olomorfa } \Rightarrow f(z) = r^n (\cos nt + j \sin nt)$$

$$\begin{cases} u = r^n \cos nt \\ v = r^n \sin nt \end{cases}$$

$$P(x, y) = x^n - C_n^2 x^{n-2} y^2 + C_n^4 x^{n-4} y^4 - \dots = \operatorname{Re} z^n$$

$\Delta P = 0$ (să se arate că este funcție armonică)

$\Rightarrow$ Partea reală a unei funcții olomorfe este armonică.

$$f^{(n)}(a) = \frac{n!}{2\pi j} \oint_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz$$

Integrale în complex

$$1. \int_{\gamma} f(z) dz$$

$F(z, \bar{z})$ - olomorfa

$$\Leftrightarrow \frac{\partial F}{\partial \bar{z}} = 0$$

(γ): $|z|=1$ (ceroul unitate)

$$f(z) = \bar{z}^2 + 2z + j\bar{z}$$

$$z = re^{jt} \text{ dar } |z|=1 \Rightarrow z = e^{jt}; t \in [0, 2\pi)$$

$$\Rightarrow f(z) = e^{-2jt} + 2e^{jt} + je^{-jt}$$

$$\bar{z} = e^{-jt}$$

$$dz = j \cdot e^{jt} dt$$

$$\int_{\gamma} f(z) dz = \int_0^{2\pi} (e^{-2jt} + 2e^{jt} + e^{-jt}) j e^{jt} dt$$

$$= j \int_0^{2\pi} (e^{-jt} + 2e^{jt} + 1) dt$$

$$= j \left[-je^{-jt} + 2e^{jt} + jt \right] \Big|_0^{2\pi} = -2\pi$$

$$e^{-jt} = \cos t - j \sin t$$

! Dacă funcția este olomorfa, integrala este zero

$$2. \int_{AB} (2z + j) e^{-5z} dz$$

$\widehat{AB}$ - arc între $A(0); B(j)$

$$f(z) = (2z + j) e^{-5z} \quad - \text{olomorfa pt ca } \frac{\partial f}{\partial \bar{z}} = 0$$

$\hookrightarrow$ nu depinde de drum

$$\Rightarrow \int_0^j (2z + j) e^{-5z} dz = \int_0^j 2z e^{-5z} dz + j \int_0^j e^{-5z} dz$$

$$1 = \cancel{2} \Rightarrow 1 = 1$$

$$g' = \frac{d}{dz} e^{-5z} = -\frac{e^{-5z}}{5}$$

$$= \left[-\frac{ze^{-5z}}{5} \right]_0^j + \frac{2}{5} \int_0^j e^{-5z} dz + j \int_0^j e^{-5z} dz$$

$$= -\frac{je^{-5j}}{5} + \left(\frac{2}{5} + j \right) \int_0^j e^{-5z} dz$$

$$= -\frac{je^{-5j}}{5} + \left(\frac{2}{5} + j \right) \left[-\frac{e^{-5z}}{5} \right]_0^j = -\frac{je^{-5j}}{5} + \left(\frac{2}{5} + j \right) \left(-\frac{e^{-5j}}{5} + \frac{1}{5} \right)$$

$$= \frac{-5je^{-5j} - 2e^{-5j} + 2 - je^{-5j} + j}{25}$$

$$= \frac{e^{-5j}(-6j-2) + 2+j}{25}$$

$$3. \int_{\gamma} f(z) dz = \int_{[AB]} f(z) dz + \int_{\widehat{BC}} f(z) dz$$

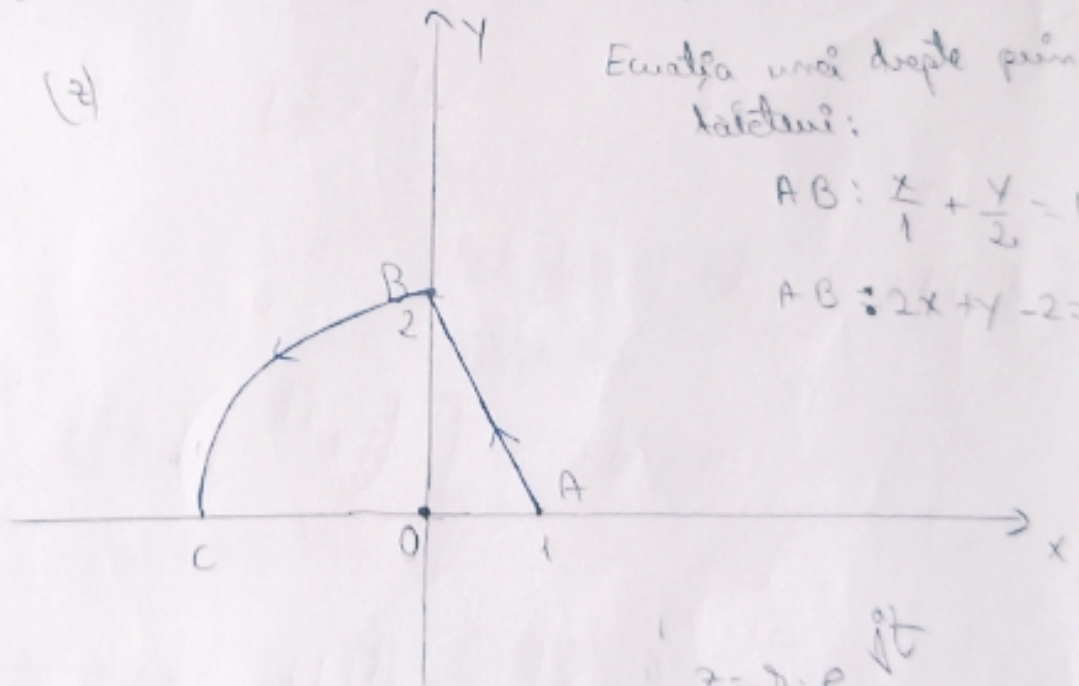
$$f(z) = \operatorname{Re} z + j \operatorname{Im} (z+j)$$

$$\gamma = [AB] \cup \widehat{BC}$$

$[AB]$ - segmentul între $A(1)$ și $B(2j)$

$\widehat{BC}$ - arc de cerc de extremități $B(2j)$; $C(-2)$

(2)



Ecuația unei drepte prin
punctele:

$$AB: \frac{x}{1} + \frac{y}{2} = 1$$

$$AB: 2x + y - 2 = 0.$$

$$f(z) = x + j(y+1)$$

$$\int_{[AB]} [x + j(y+1)] dz =$$

$$z = x + jy$$

$$z|_{AB} = x + 2j(1-y)$$

$$dz|_{AB} = dx - 2j dx = (1-2j) dx$$

$$z = r \cdot e^{jt} \\ = 2e^{jt}; t \in \left(\frac{\pi}{2}, \pi\right)$$

$$\begin{aligned}
\int_0^1 [x + j \cdot 2(1-x) + j] \cdot (1-2j) dx &= (1-2j) \int_0^1 (x + 2j - xj + j) dx \\
&= (2j-1) \int_0^1 [x(1-j) + 3j] dx = (2j-1) \left[\frac{x^2(1-j)}{2} + 3xj \right]_0^1 \\
&= (2j-1) \left[\frac{1-j}{2} + 3j \right] \\
&= j(1-j) - \frac{1-j}{2} - 6 - 3j \\
&= j + 1 - \frac{(1-j)}{2} - 6 - 3j \\
&= -5 - 2j - \frac{1-j}{2} \\
&= -\frac{3}{2} - \frac{1}{2} - 2j + \frac{1}{2}j \\
&= -\frac{11}{2} - \frac{3}{2}j
\end{aligned}$$

$$\begin{aligned}
\int_{\overline{BC}} f(z) dz &= \int_{\overline{BC}} [x + j(y+1)] dz \\
&= \int_{\frac{\pi}{2}}^{\pi} (z + j) dz = \int_{\frac{\pi}{2}}^{\pi} (2e^{it} + j) 2je^{jt} dt \\
&= 2j \int_{\frac{\pi}{2}}^{\pi} (2e^{2it} + j e^{2jt}) dt \\
&= 2j(2+j) \int_{\frac{\pi}{2}}^{\pi} e^{2jt} dt \\
&= 2j(2+j) \left[\frac{e^{2jt}}{2j} \right]_{\frac{\pi}{2}}^{\pi} = (2+j)(e^{2\pi j} - e^{j\pi}) = 2(2+j)
\end{aligned}$$

$$e^{2k\pi j} = 1 \quad \forall k \in \mathbb{Z}$$

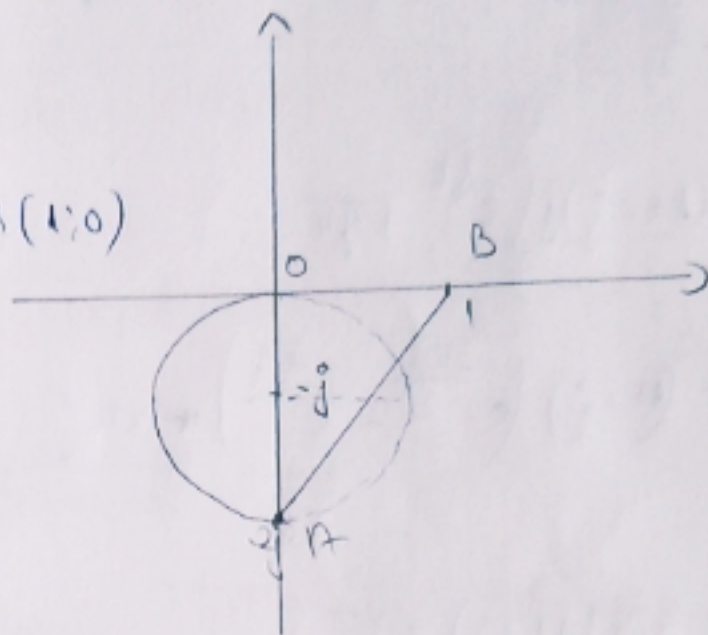
Serie Taylor, Serii Laurent

Ex 1 Sa se calculeze:

$$I = \int_{\gamma} [(z+i) + 2\bar{z}] dz \quad \text{unde } \gamma$$

$$\gamma = \widehat{OA} \cup \overline{AB}$$

$$OA \subseteq \Gamma(-i; 1); B(1; 0)$$



$$I = \int_{\widehat{OA}} [(z+i) + 2\bar{z}] dz + \int_{\overline{AB}} [(z+i) + 2\bar{z}] dz$$

$$(\gamma) = \begin{cases} x = x(t) \\ y = y(t) \Leftrightarrow z = z(t); \quad 0 \leq t \leq b \end{cases}$$

$$\int_{\gamma} f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

$$I_1; \widehat{OA} \subseteq \Gamma(-i; 1)$$

$$z = -i + 1 \cdot e^{it}; \quad \frac{\pi}{2} \leq t \leq \frac{3\pi}{2}$$

$$\Gamma(z_0; r); z = z_0 + r e^{it} \quad 0 \leq t \leq 2\pi$$

$$dz = j e^{jt} dt$$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |f(re^{it}, j)| + 2(j + e^{-it}) \cdot j \cdot e^{jt} dt$$

$$|e^{jt} \cdot j| = 1$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + 2j + 2e^{-it}) j e^{jt} dt$$

$$= (1 + 2j) j \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{jt} dt + 2j \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{-it} dt$$

$$= (1 + 2j) \left(e^{\frac{3\pi}{2}j} - e^{\frac{\pi}{2}j} \right) + 2j \left(e^{\frac{\pi}{2}j} - e^{-\frac{\pi}{2}j} \right)$$

$$= -2j(1 + 2j) + 2\pi j$$

$$= 2(\pi j - j + 2)$$

$$A(0, -2)$$

$$B(1, 0)$$

$$AB: \frac{x-0}{1-0} = \frac{y+2}{0+2} = t$$

$$\Rightarrow \begin{cases} x = t \\ y = 2t - 2 \end{cases} \quad 0 \leq t \leq 1$$

$$x + jy = t + 2jt - 2j, t \in [0, 1]$$

$$z = t(1 + 2j) - 2j$$

$$dz = (1+2j) dt$$

$$i_2 = \int_0^1 |t(1+2j) - 2j + j| + 2[t(1-2j) + 2j] dt$$

$$= (1+2j) \int_0^1 |t + j(2t-1)| dt + 2(1+2j) \int_0^1 [t(1-2j) + 2j] dt$$

$$= (1+2j) \int_0^1 \sqrt{t^2 + (2t-1)^2} dt + 2(1+2j) \left[(1-2j) \frac{t^2}{2} + 2jt \right]_0^1$$

$$= (1+2j) \int_0^1 \sqrt{5t^2 - 4t + 1} dt + 2(1+2j) \left[\frac{(1-2j)}{2} + 2j \right]$$

met ou $f = \int_0^1 \sqrt{5t^2 - 4t + 1} dt$

metoda 1: Euler $\sqrt{5t^2 - 4t + 1} = t\sqrt{5} + u$

metoda 2: $\sqrt{5t^2 - 4t + 1} = tu + 1$

metoda 3: $\sqrt{5t^2 - 4t + 1} = \sqrt{5 \left[\left(t - \frac{2}{5}\right)^2 + \frac{1}{25} \right]}$

$$= \sqrt{5} \cdot \sqrt{u^2 + \frac{1}{25}}$$

$$\int \sqrt{u^2 + \frac{1}{25}} du = \int \frac{u^2 + \frac{1}{25}}{\sqrt{u^2 + \frac{1}{25}}} du = \int u \cdot \frac{u}{\sqrt{u^2 + \frac{1}{25}}} du + \frac{1}{25} \int \frac{1}{\sqrt{u^2 + \frac{1}{25}}} du$$

$$\downarrow$$

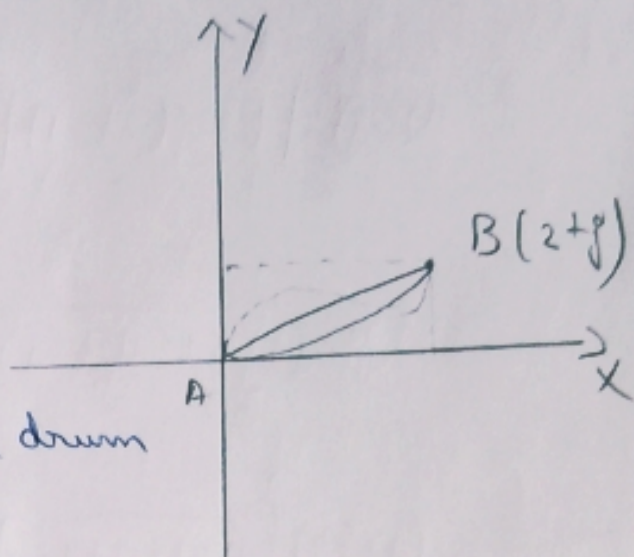
$$u \left(\sqrt{u^2 + \frac{1}{25}} \right)'$$

Ex 2 $i = \int_{\widehat{AB}} (2z + 3j) e^{jz} dz$

unde: $A(0)$; $B(2+j)$

$$\begin{cases} f(z) = (2z + 3j) e^{jz} \\ \text{alemamorfa în } \mathbb{C} \Rightarrow \end{cases}$$

i nu depinde de drum



$$\frac{\partial f}{\partial \bar{z}} = 0$$

$$F(z) = \int f(z) dz = \int (2z + 3j) e^{jz} dz =$$

$$= \int (2z + 3j) \left(\frac{e^{jz}}{j} \right)' dz = \frac{2z + 3j}{j} e^{jz} - \int \frac{2}{j} e^{jz} dz$$

$$= (3 - 2jz) e^{jz} + 2 e^{jz}$$

$$i = F(z) \Big|_A^B = F(B) - F(A) = [3 - 2j(2+j)] e^{j(2+j)} +$$

$$+ 2 e^{j(2+j)} - [3 + 2] = (5 - 4j) e^{j-1} + 2 e^{2j-1} - 5 =$$

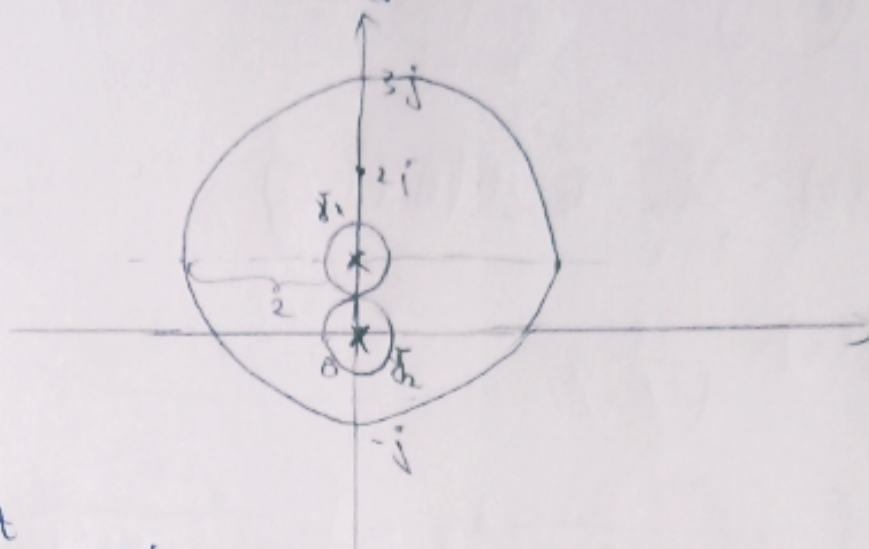
$$= e^{2j-1} (2 - 4j) - 5$$

$$e^{2j-1} = e^{2j} \cdot e^{-1} = \frac{1}{e} (\cos 2 + j \sin 2)$$

$$2 \in \left[\frac{\pi}{2}, \pi \right]$$

Ex 3 $\int \frac{e^{2z} + 3jz}{3(z-j)^2(z+2)} dz$
 $|z-j|=2$

(8): $|z-j|=2 \Rightarrow z \in \Gamma(j; 2)$



$z = j + 2e^{jt}; 0 \leq t \leq 2\pi$

$|z|^2 = z \cdot \bar{z}$

$g(z) = \frac{e^{2z} + 3jz}{3(z-j)^2(z+2)}$

$\begin{cases} z_1 = 0 \rightarrow \text{pole simple} \\ z_2 = j \rightarrow \text{pole double} \\ z_3 = -2 \rightarrow \text{pole triple} \end{cases}$

T Cauchy $\left. \begin{array}{l} f \text{ - analyt. in int}(\gamma) \\ f \text{ - cont. pe}(\gamma) \end{array} \right\} \Rightarrow \int_{\gamma} f(z) dz = 0$

$f(z_0) = \frac{1}{2\pi j} \int_{\gamma} \frac{f(z)}{z-z_0} dz \Rightarrow \int_{\gamma} \frac{f(z)}{z-z_0} dz = 2\pi j f(z_0) \quad (1)$

$f^{(n)}(z_0) = \frac{n!}{2\pi j} \int_{\gamma} \frac{f(z)}{(z-z_0)^{n+1}} dz \Rightarrow \int_{\gamma} \frac{f(z)}{(z-z_0)^{n+1}} dz = \frac{2\pi j}{n!} f^{(n)}(z_0)$

$$I = \int_{\gamma} \frac{e^{2z}}{z(z-2j)(z+2)} dz + \int_{\gamma} \frac{3j}{(z-2j)^2(z+2)} dz = I_1 + I_2$$

$$I_2 = \int_{\gamma} \frac{3j}{(z-2j)^2} dz \quad \begin{matrix} \rightarrow f(z) \\ (2) \\ n=1 \\ z_0=2j \end{matrix} \quad \frac{2\pi j}{1!} f'(z_0) = \frac{6\pi}{8j} = -\frac{3\pi j}{4}$$

$$f(z) = \frac{3j}{z+2} \in \mathcal{H}(\overline{D}(j; 2))$$

$$f'(z) = \frac{-3j}{(z+2)^2}$$

$$I_1 = \underbrace{\int_{\gamma_1} \frac{e^{2z}}{z(z-2j)^2(z+2)} dz}_{I_3} + \underbrace{\int_{\gamma_2} \frac{e^{2z}}{z(z-2j)^2(z+2)} dz}_{I_4}$$

$$I_3 = \int_{\gamma_1} \frac{e^{2z}}{z(z-2j)^2(z+2)} dz \stackrel{(1)}{=} 2\pi j \frac{e^{2z}}{(z-2j)^2(z+2)} \Big|_{z=0}$$

$$= 2\pi j \cdot \frac{1}{-4 \cdot 2} = \frac{2\pi j}{-8} = -\frac{\pi j}{4}$$

$$I_4 = \int_{\gamma_2} \frac{e^{2z}}{(z-2j)^2} dz \quad \begin{matrix} (2) \\ z_0=2j \\ n=1 \end{matrix} \quad \frac{2\pi j}{1!} \left(\frac{e^{2z}}{z(z+2)} \right)' \Big|_{z=2j}$$

$$= \frac{2\pi j}{1!} \cdot \frac{2e^{2z}z(z+2) - e^{2z}(1+z)}{z^2(z+2)^2} \Big|_{z=2j} =$$

$$= \frac{e^{2z} [2z(z+2) - 2z - 2]}{z^2(z+2)^2} = \frac{e^{2z} (2z^2 + 2z - 2)}{z^2(z+2)^2} \Big|_{z=2j}$$

$$= \frac{e^{4j} (-8 + 4j - 2)}{-4(2j+2)^2} \cdot 2j - \frac{2e^{4j} (5 - 2j)}{-32j} \cdot 2\pi j$$

$$= \frac{e^{4j} (2-5j)}{16} \cdot 2\pi j = \frac{e^{4j} (2j+5)}{8} \pi$$

$$i = -\frac{\pi j}{4} + \frac{e^{4j} (2j+5)}{8} \pi - \frac{3\pi j}{4}$$

$$= \pi \left[\frac{e^{4j} (2j+5)}{8} - j \right]$$

Ex 4 $f: \mathbb{C} \setminus \{j; 1\} \rightarrow \mathbb{C}$

$$f(z) = \frac{2z+1}{(z-j)(z-1)^2} = \frac{A}{z-j} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

Să se dezvoltă f în serie în jurul punctelor: a) $z_0 = 0$
b) $z_0 = 1$

c) Să se precizeze natura punctelor

$$z_0 = 0$$

$$z_0 = 1$$

$$z_0 = j$$

$$A(z-1)^2 + B(z-1)(z-j) + C(z-j) = 2z+1$$

$$z=j \Rightarrow 2j+1 = A(j-1)^2 \Rightarrow A = \frac{2j+1}{-2j} = -\frac{2j}{2}$$

$$z=1 \Rightarrow 3 = C(1-j) \Rightarrow C = \frac{3}{(1-j)} = \frac{3(1+j)}{2}$$

$$A+B=0$$

$$\Rightarrow B = -A = \frac{2-j}{2}$$

$$f(z) = \frac{-2+j}{2} \cdot \frac{1}{z-j} + \frac{2-j}{2} \cdot \frac{1}{z-1} + \frac{3(1+j)}{2} \cdot \frac{1}{(z-1)^2}$$

$$\frac{1}{z-1} = -\frac{1}{1-z} = -\sum_{n=0}^{\infty} z^n ; |z| < 1 \quad \left| (1) \right| \cdot \frac{2-j}{2}$$

$$-\frac{1}{(z-1)^2} = -\sum_{n=1}^{\infty} n \cdot z^{n-1}$$

$$\Rightarrow \frac{1}{(z-1)^2} = \sum_{n=0}^{\infty} (n+1) z^n \quad \left| \frac{3(1+j)}{2} \right|$$

$$\frac{1}{z-j} = \frac{1}{-j} \cdot \frac{1}{1-\frac{z}{j}} = j \cdot \frac{1}{1-\frac{z}{j}} = j \sum_{n=0}^{\infty} \left(-1\right)^n \frac{1}{j} \cdot z^n \quad \left| \cdot \frac{j-2}{2} \right|$$

$$f(z) = \sum_{n=0}^{\infty} \left[\frac{j-2}{2} + \frac{3}{2} (1+j)(n+1) + (-1)^{n+1} \frac{1}{j} \cdot \frac{4+2j}{2} \right] z^n$$

$\underbrace{\hspace{15em}}_{a_n}$

$$= \sum_{n=0}^{\infty} a_n z^n$$

Seria obținută este o serie Taylor dea p.p. a seriei Laurent este nulă (unde p.p - parte principală)
 $\Rightarrow z_0 = 0$ - punct ordinar (nu e ind. a punctului)

$$2) z - z_0 = u \Leftrightarrow z - 1 = u$$

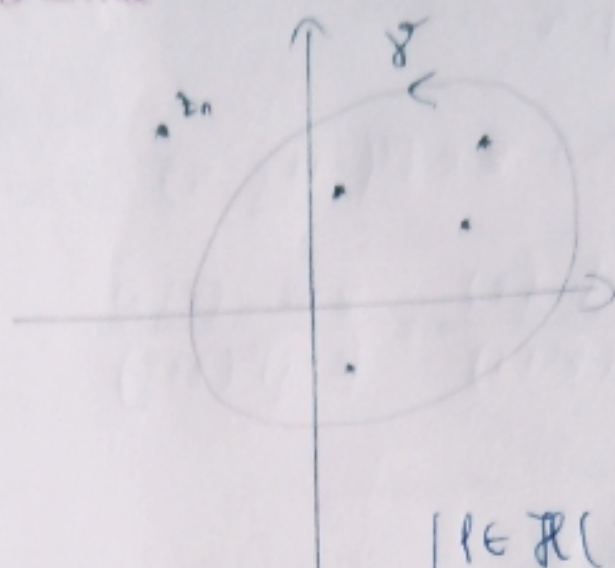
$$z = u + 1$$

$$f(z) = \frac{j-2}{2} \cdot \frac{1}{u+1-j} + \underbrace{\frac{2-j}{2} \cdot \frac{1}{u} + \frac{3(1+j)}{2} \cdot \frac{1}{u}}_{\text{P.P.}}$$

$z_0 = 1$ - pol double

$z_0 = j$ - pol simple

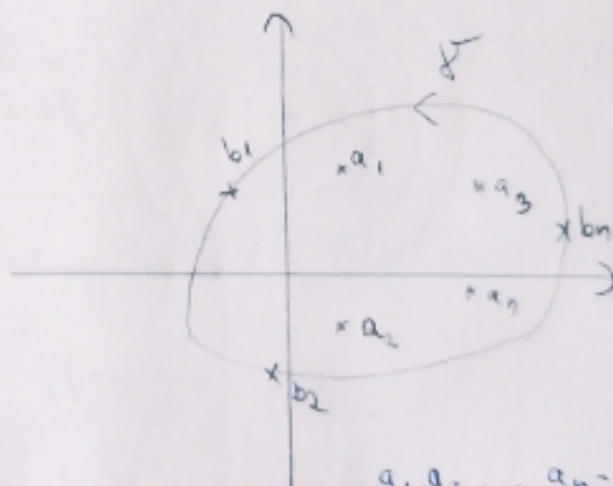
Teorema Rezidurilor



$$\begin{cases} f \in \mathcal{H}(\text{Int}(\gamma)) \\ f \in C(\gamma) \end{cases}$$

$$z_k \in \text{Int}(\gamma); 1 \leq k \leq n$$

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$



$a_1, a_2, \dots, a_n$ - puncte singulare
 $\in \text{Int}(\gamma)$

$b_1, b_2, \dots, b_n$ - poli simple
 $\in \gamma$

Teorema semirezidurilor

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, a_k) + \pi i \sum_{k=1}^m \text{Res}(f, b_k)$$

Ex 1 Sa se calculeze

$$I(n) = \int \frac{z \cdot \sin(\pi j z)}{(z-j)(z^2+1)} dz; \quad n > 0; n \neq 1$$

$$|z+1-j| = n$$

$$|z+1-j| = n \Leftrightarrow z \in \Gamma(-1+j; n)$$

$$f(z) = \frac{z \cdot \sin(\pi j z)}{(z-j)(z^2+1)} = \frac{z \sin(\pi j z)}{(z-j)^2(z+j)}$$



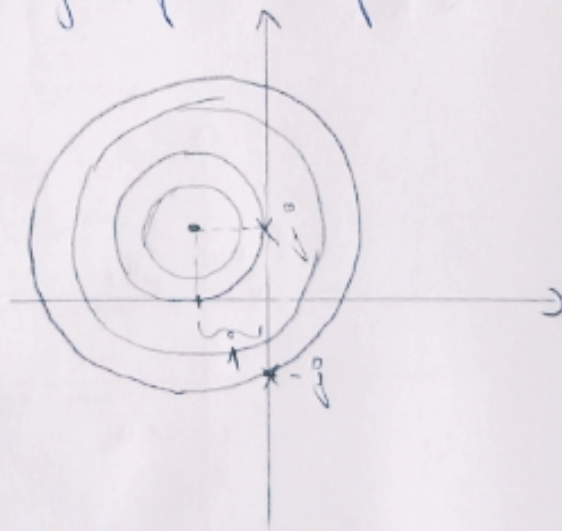
functie meromorfa

rađacinski va j doar cele ale numitorului

$$z^2+1 = z^2+j^2 = (z-j)(z+j)$$

$$(z-j)^2 = 0 \Rightarrow z_1 = j - \text{pol dublu}$$

$$z+j=0 \Rightarrow z_2 = -j - \text{pol simplu}$$



I $n \in (0, 1)$

$$\int_{\gamma} f(z) dz = 2\pi j \sum_{k=1}^{\infty} \text{Res}(f, z_k) = 0$$

$$I(n) = 0$$

II $n=1$ - exclus din date problema
(integrala divergenta)

$$\text{III} \quad = |z_2 - z_0| = |-j + 1-j| = |1-2j| = \sqrt{5}$$

$$n \in (1, \sqrt{5}) \Rightarrow i(n) = 2\pi j \operatorname{Res}(f, j)$$

$$\begin{cases} j \in \operatorname{Int}(\gamma) \\ -j \in \operatorname{Ext}(\gamma) \end{cases}$$

$z = z_0$ - pol de ordin n

$$\Rightarrow \operatorname{Res}(f; z_0) = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} (z - z_0)^n \cdot f'(z)$$

$$\text{IV} \quad n = \sqrt{5}$$

$$\begin{cases} j \in \operatorname{Int}(\gamma) \\ -j \in \operatorname{Ext}(\gamma) \end{cases}$$

$n=1$
(pol simplu) $z \rightarrow z_0$

$$\Rightarrow i(n) = 2\pi j \operatorname{Res}(f, j) + \pi j \operatorname{Res}(f, -j)$$

$\uparrow$
Teorema reziduurilor

$$\text{V} \quad n > \sqrt{5}$$

$$\Rightarrow \pm j \in \operatorname{Int}(\gamma)$$

$$i(n) = 2\pi j (\operatorname{Res}(f, j) + \operatorname{Res}(f, -j))$$

$$\operatorname{Res}(f, -j) = \lim_{z \rightarrow -j} \frac{(z+j) \cdot z \cdot \sin \pi j z}{(z-j)^2 (z+j)} = \frac{-j \sin \pi}{-4} = 0$$

Obs $z = -j$ - singularitate eliminabilă

$$\operatorname{Res}(f, j) = \lim_{z \rightarrow j} \left[\frac{(z-j)^2 \cdot z \cdot \sin \pi j z}{(z-j)^2 (z+j)} \right]'$$

$$= \lim_{z \rightarrow j} \left[\frac{z \sin(2\pi j)}{z+j} \right]'$$

$$= \lim_{z \rightarrow j} \frac{\{ \sin(2\pi j) + z \cos(2\pi j) \} (z+j) - z \sin \pi j z}{(z+j)^2}$$

$$= \frac{2j\pi}{1} - j\frac{\pi}{2}$$

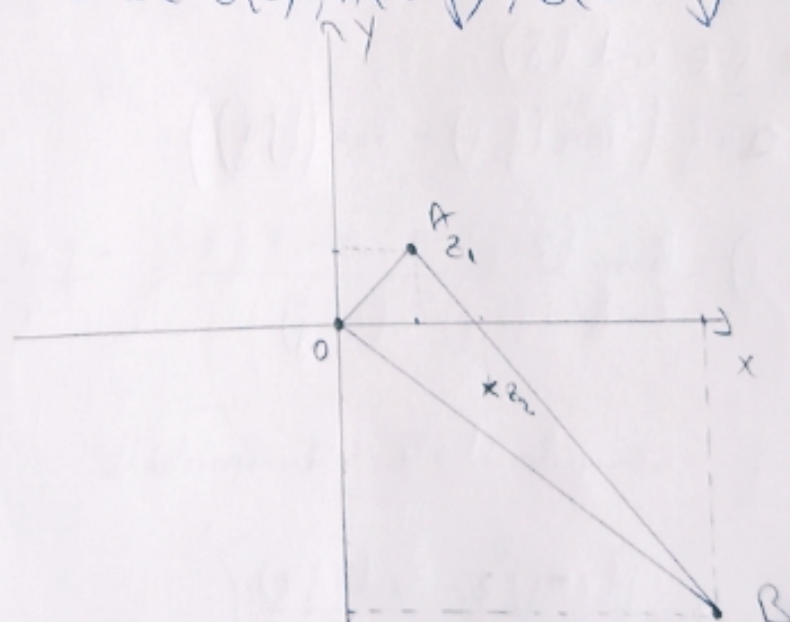
$$i(z) = \begin{cases} 0 & ; z \in (0, 1) \\ \frac{2\pi j}{1} - \frac{j\pi}{2} = \pi^2 & ; z \in (1, \sqrt{5}) \\ \pi^2 & ; z = \sqrt{5} \\ 2\pi^2 & ; z > \sqrt{5} \end{cases}$$

Ex 2 São se calculare

$$i = \int_T \frac{2z+1}{z^2-3z+3+j} dz$$

unde

$$T = \triangle OAB \text{ unde } O(0); A(1+j); B(5-4j)$$



$$f(z) = \frac{2z+1}{z^2-3z+3+j}$$

$$z^2-3z+3+j=0$$

$$\Delta = 9 - 4 \cdot 1(3+j) = 9 - 12 - 4j = -3 - 4j = 4j^2 + 1 - 4j = (2j-1)^2$$

$$z_{1,2} = \frac{3 \pm (2j-1)}{2}$$

$$z_1 = \frac{2+2j}{2} = 1+j \in \gamma$$

$$z_2 = \frac{4+2j}{2} = 2-j \in \text{Int}(\gamma)$$

$$i = 2\pi j \text{Res}(f; 2-j) + \pi j \text{Res}(f; 1+j)$$

$$f(z) = \frac{g(z)}{h(z)} \quad \text{si } z_0 \text{ - pol simplu}$$

$$\Rightarrow \begin{cases} g(z_0) \neq 0 \\ h(z_0) = 0 \\ h'(z_0) \neq 0 \end{cases} \Rightarrow \text{Res}(f; z_0) = \left. \frac{g(z)}{h'(z)} \right|_{z=z_0}$$

Metoda 1

$$\text{Res}(f; 2-j) \Rightarrow \left. \frac{2z+1}{(2z-3)} \right|_{z=2-j}$$

$$\approx \left. \frac{2z+1}{(z^2-3z+3-j)'} \right|_{z=2-j}$$

$$= \frac{5-2j}{1-2j}$$

$$= \frac{(5-2j)(1+2j)}{1+4} = \frac{5+10j-2j+4}{5}$$

$$= \frac{9+8j}{5}$$

Metoda 2

$$\text{Res}(f; 1+j) = \lim_{z \rightarrow 1+j} (z-1-j) \cdot \frac{2z+1}{(z-1-j)(z-2+j)} dz$$

$$\lim_{z \rightarrow 1+j} \frac{z+1}{(z-2+j)^2} = \frac{1+j+2}{-1+j} = \frac{1-8j}{5}$$

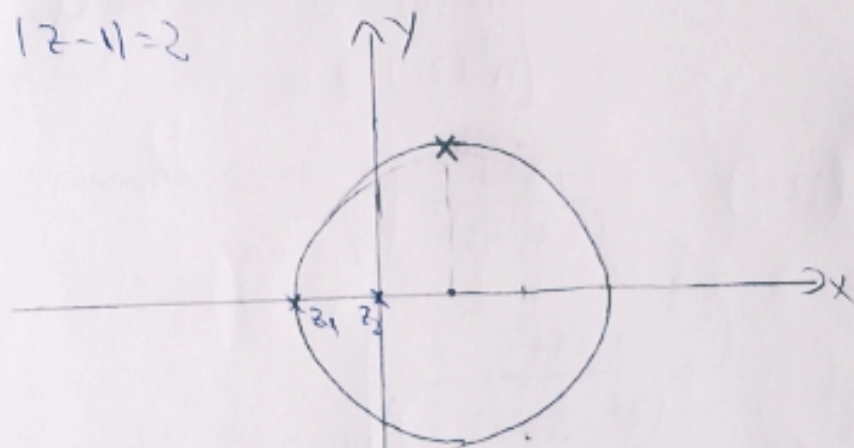
$$i = \pi j \left(\frac{18+16j}{5} + \frac{1-8j}{5} \right)$$

$$= \frac{\pi j}{5} (18+16j+1-8j)$$

$$= \frac{\pi j}{5} (8j+19)$$

Ex 3 $i = \int \frac{\operatorname{sh} \frac{j}{z}}{z+1} dz$ $z \in \Gamma(1; 2)$

$|z-1|=2$



$$z+1=0$$

$z = -1$ - punct simple (pol)

$z=0$ - punct singular essential e Int(f)

$$i = 2\pi j \operatorname{Res}(f; 0) + \pi j \operatorname{Res}(f; -1)$$

$$\operatorname{Res}(f; -1) = \lim_{z \rightarrow -1} (z+1) f(z)$$

$$= \lim_{z \rightarrow -1} (z+1) \frac{\operatorname{sh} \frac{j}{z}}{z+1}$$

$$= \operatorname{sh} \frac{j}{-1} = \operatorname{sh}(-j) = -j \sin 1$$

$\text{Res}(f; 0) = \text{coefficiente bei } \frac{1}{z-20}$ der serie Laurent

$$\frac{1}{z+1} = 1 - z + z^2 - z^3 + \dots$$

$$\sinh z = \frac{1}{1!} z + \frac{1}{3!} z^3 + \dots \quad \text{at } z=j$$

$$\Rightarrow \sinh \frac{j}{z} = \frac{1}{1!} \cdot \frac{j}{z} + \frac{1}{3!} \cdot \frac{j^3}{z^3} + \frac{1}{5!} \cdot \frac{j^5}{z^5} + \dots$$

$$\text{Res}(f; 0) = \frac{1}{1!} \cdot j + \frac{1}{3!} \cdot \frac{j^3}{z^3} + \frac{1}{5!} \cdot \frac{j^5}{z^5}$$

$$\Rightarrow \sinh j = \frac{1}{1!} j + \frac{1}{3!} j^3 + \frac{1}{5!} j^5 + \dots$$

$$\Rightarrow \text{Res}(f; 0) = \sinh j = j \cdot \sin 1$$

$$\Rightarrow j = 2\pi j(j \sin 1) + \pi j(-j \sin 1)$$

$$= j \sin 1 (2\pi j - \pi j)$$

$$= -\pi \sin 1$$

Obs

$$\text{H.2 } \text{Res}(f; 0) + \text{Res}(f; -1) + \text{Res}(f; \infty) = 0$$

$$\text{Res}(f; \infty) = -\text{Res}\left[\frac{1}{z^2} \cdot f\left(\frac{1}{z}\right); 0\right]$$

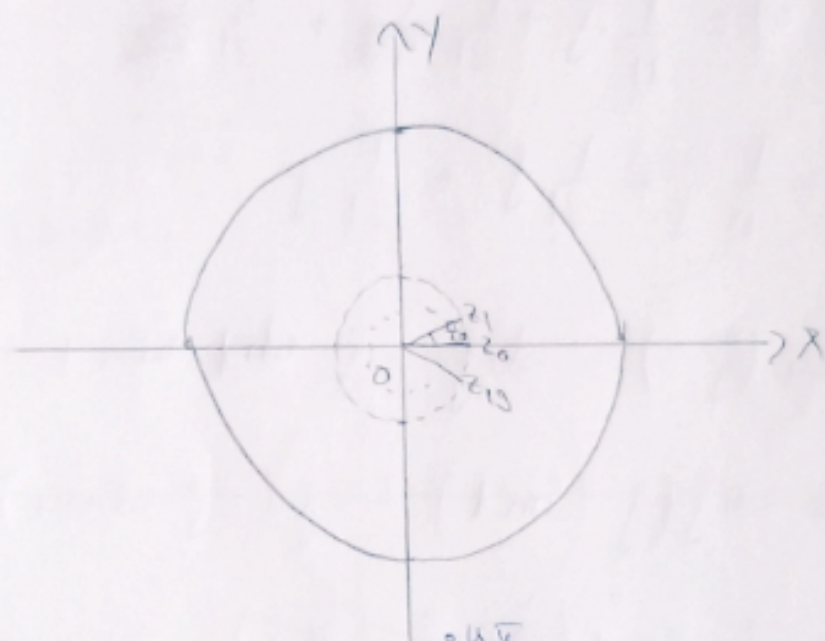
$$= -\text{Res}\left[\frac{1}{z^2} \cdot \frac{\sinh z}{\frac{1}{z} + 1}; 0\right]$$

$$z = \text{Res} \left(\frac{\sinh z}{z(z+1)}; 0 \right)$$

$$= \lim_{z \rightarrow 0} z \cdot \frac{\sinh z}{z(z+1)} = \frac{\sinh 0}{1} = 0$$

$$\text{Res}(f; 0) = -\text{Res}(f; -1) = j \sin 1$$

Ex 4 $i = \int_{|z|=3} \frac{\cos \frac{j}{z}}{z^3 (z^{20} - 1)} dz$



$$z^{20} - 1 = 0$$

$$z_k = \cos \frac{2k\pi}{20} + j \sin \frac{2k\pi}{20} = e^{j \frac{k\pi}{10}}; 0 \leq k \leq 19$$

$z_{20} = 0$ - point singular essential

$$i = 2\pi j \sum_{k=0}^{20} \text{Res}(f; z_k) \quad (1)$$

$$\textcircled{T} \sum_{k=0}^{20} \text{Res}(f; z_k) + \text{Res}(f; \infty) = 0 \quad (2)$$

$$\text{Res}(f; \infty) = -\text{Res} \left(\frac{1}{z} f \left(\frac{1}{z} \right); 0 \right)$$

$$= -\operatorname{Res} \left[\frac{1}{z^3} \cdot \frac{\cos(jz)}{\frac{1}{z^3} \left(\frac{1}{z^{20}} - 1 \right)} ; 0 \right]$$

$$= -\operatorname{Res} \left[\frac{1}{z^2} \cdot \frac{\cos(jz)}{1 - z^{20}} ; 0 \right]$$

$$= -\operatorname{Res} \left[\frac{z^{21} \cdot \cos(jz)}{1 - z^{20}} ; 0 \right]$$

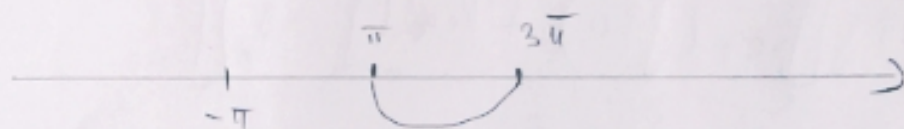
$z=0$ - point ordinar

$$\Rightarrow \operatorname{Res}(f; \infty) = 0$$

$$\Rightarrow \operatorname{Res}(1) \operatorname{Res}(2) \quad \text{ca } i = -2\pi j \cdot \operatorname{Res}(f; \infty) = 0$$

$$\text{Ex 5} \quad i = \int_{-\pi}^{3\pi} \frac{2 + \cos x}{7 \sin x + 25} dx$$

$$T = 2\pi$$



$$i = 2 \int_{-\pi}^{\pi} \dots = 2 \int_0^{2\pi} \frac{2 + \cos x}{7 \sin x + 25} dx$$

$$e^{jx} = z \quad ; x \in [0; 2\pi] \Rightarrow z \in \Gamma(0; 1)$$

$|z| = 1$

$$\begin{cases} \sin x = \frac{z^2 - 1}{2jz} \\ \cos x = \frac{z^2 + 1}{2z} \end{cases}$$

$$\int_0^{2\pi} e^{jx} dx = dz \Rightarrow dx = \frac{dz}{jz}$$

$$j = 2 \int_{|z|=1} 2 + \frac{z^2 - 1}{2jz} \cdot 7 \cdot \frac{z^2 - 1}{2jz} + 25 \cdot \frac{dz}{jz}$$

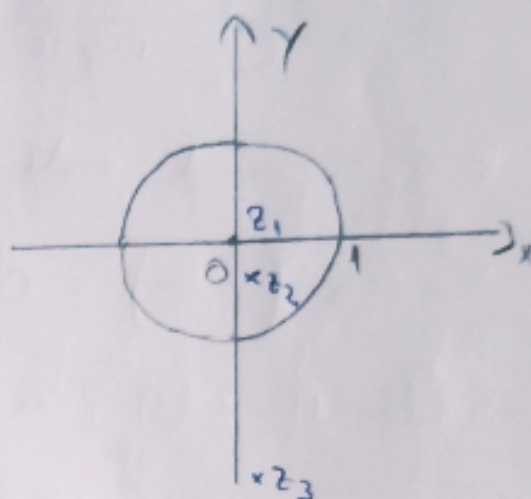
$$= 2 \int_{|z|=1} \frac{z^2 + 1 + 4z}{z(7z^2 + 50jz - 7)} dz$$

$$z_1 = 0$$

$$\Delta = 48j^2$$

$$\Rightarrow z_{2,3} = \frac{-50j \pm 48j}{14}$$

$$\left. \begin{aligned} z_2 &= -\frac{j}{7} \\ z_3 &= -7j \end{aligned} \right\} \text{poli simple}$$



Tema

$$j = 2 \cdot 2\pi j \left[\text{Res}(f; 0) + \text{Res}(f; -\frac{j}{7}) \right]$$

Rezultatul final trebuie să fie un nr. real pos.

Seminar 7
8. aprilie 2014

$$\int_a^{a+2\pi} R(\sin x, \cos x) dx = \int_a^{a+2\pi} R(\sin u, \cos u) du$$

$u = \text{unghi}$

$$e^{jx} = z$$

$$\cos x = \frac{e^{jx} + e^{-jx}}{2} = \frac{z^2 + 1}{2z}$$

$$\sin x = \frac{z^2 - 1}{2jz}$$

$x = \text{unghi}$

$$\int_a^{a+2\pi}$$

$$= \int_a^{a+2\pi}$$

$$+ \int_{a+2\pi}^{a+4\pi}$$

$$= 2 \cdot \int_a^{a+2\pi}$$

$$\int_{-\pi}^{\pi}$$

$$a = -\pi$$

$$\int e^{jx} dx = dz$$

$$dx = \frac{dz}{jz}$$

Exerciții

1.

$$\int_0^{2\pi}$$

$$\frac{1 + \sin x}{2 - \cos x} dx$$

$$= \int_{|z|=1}$$

$$\frac{1 + \frac{z^2 - 1}{2jz}}{2 - \frac{z^2 + 1}{2z}} \cdot \frac{1}{jz} dz$$

$$\left. \begin{array}{l} e^{jx} = z \\ |z|=1 \end{array} \right\}$$

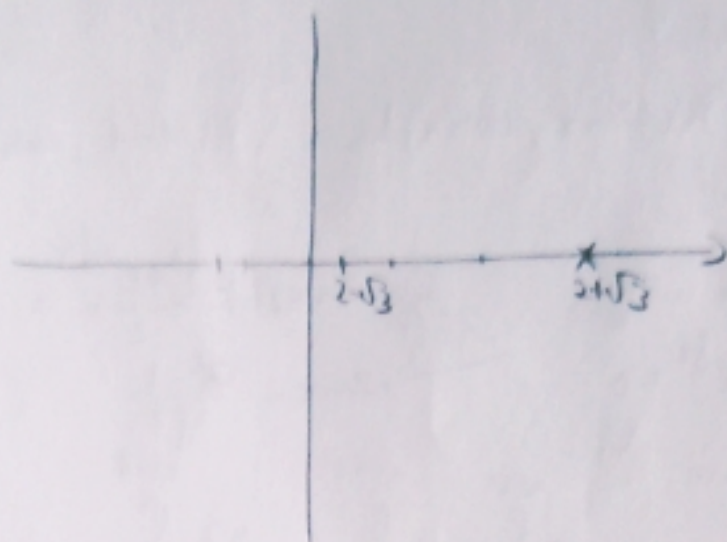
$$= \int_{|z|=1} \frac{z^2 + 2jz - 1}{2jz} \cdot \frac{z}{z^2 - 1} \cdot \frac{1}{jz} dz$$

$$= \int_{|z|=1} \frac{(z + j)^2}{z(z^2 - 4z + 1)} dz$$

$$z(z^2 - 4z + 1) = 0$$

$$z = 0$$

$$\text{sau } (z - 2)^2 = 0 \Rightarrow z_{2,3} = 2$$



$$= 2\pi i (R_0 + R_{2-\sqrt{3}}) = \frac{2\pi i \frac{2}{-\sqrt{3}}}{-\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

o.c. $2-\sqrt{3}$ - poli simple

$\lim_{z \rightarrow z_0} \frac{g(z)}{h(z)}$
 z_0 - poli simple
 $g(z_0) \neq 0$

$$\left. \begin{array}{l} h(z_0) = 0 \\ h'(z_0) \neq 0 \end{array} \right\} \Rightarrow R_{z_0} = \frac{g(z_0)}{h'(z_0)}$$

$$\frac{z^2 + 2iz - 1}{z^2 - 4z + 11z(2z-1)}$$

$$R_0 = \frac{-1}{1} = -1$$

$$R_{2-\sqrt{3}} = \frac{4 - 4\sqrt{3} + 3 + 4i - 2i\sqrt{3} - 1}{(2-\sqrt{3})(2-\sqrt{3}-4)}$$

$$= \frac{6 - 4\sqrt{3} + i(4-2\sqrt{3})}{4 - 2\sqrt{3} - 8 - 2\sqrt{3} + 3 + 4i\sqrt{3}} = \frac{4\sqrt{3} - 6 - i(4-2\sqrt{3})}{-2\sqrt{3}(2-\sqrt{3}) - 6 - 2i(2-\sqrt{3})} = \frac{\sqrt{3}-i}{\sqrt{3}}$$

$$-2\sqrt{3}(2-\sqrt{3}) - 6 - 2i(2-\sqrt{3}) = \frac{\sqrt{3}-i}{\sqrt{3}}$$

$$I = \int_{-\pi}^{\pi} \frac{a \sin^2 nx}{p^2 - 2p \cos x + 1} dx = 2 \int_{-\pi}^{\pi} \frac{\sin^2 nx}{p^2 - 2p \cos x + 1} dx$$

$$n \in \mathbb{N}$$

$$p \in \mathbb{R} \setminus \{-1, +1\}$$

$$e^{ix} = z$$

$$2 \sin^2 x = 1 - \cos 2x$$

$$I = \int_{-\pi}^{\pi} \frac{1 - \cos 2nx}{p^2 - 2p \cos x + 1} dx$$

$$I = \int_{-\pi}^{\pi} \frac{1}{p^2 - 2p \cos x + 1} dx = \int_{|z|=1} \frac{1}{p^2 - p \cdot \frac{(z^2+1)}{z} + 1} \cdot \frac{dz}{iz}$$

$$e^{ix} = z \Rightarrow |z|=1$$

$$\cos x = \frac{z^2+1}{2z}$$

$$= \oint_{|z|=1} \frac{dz}{p^2 z - p(z^2+1) + z}$$

$$dx = \frac{dz}{iz}$$

$$= \oint_{|z|=1} \frac{dz}{p^2 z - p z^2 - p + z}$$

$$= \oint_{|z|=1} \frac{1}{p z^2 - (p^2+1)z + p} dz$$

$$\Delta = (p^2+1)^2 - 4 \cdot p \cdot p = (p^2+1-2p)(p^2+1+2p)$$

$$= (p^2 - 2p + 1)(p^2 + 2p + 1)$$

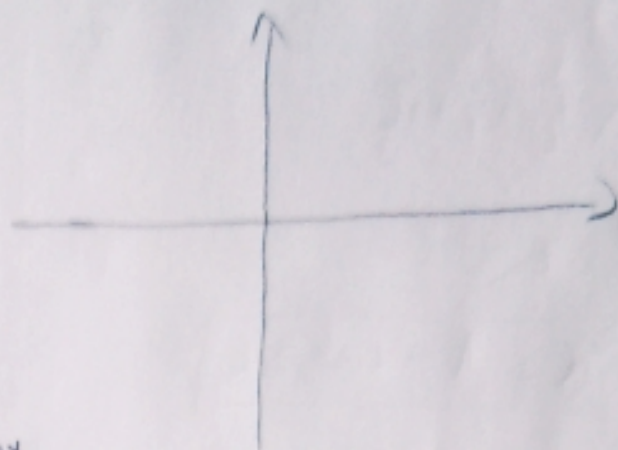
$$= (p-1)^2 (p+1)^2$$

$$\Rightarrow (p^2-1)^2$$

$$z_{1,2} = \frac{p^{2n+1} \pm |(p^2-1)|}{2p}$$

$$z_1 = p$$

$$z_2 = \frac{1}{p}$$



$$\cos 2nx = \frac{e^{j2nx} + e^{-j2nx}}{2} = \frac{z^{2n} + z^{-2n}}{2} = \frac{z^{4n} + 1}{2z}$$

$$J = \int_{-\pi}^{\pi} \frac{\cos 2nx \, dx}{p^2 - 2p \cos x + 1} + j \int_{-\pi}^{\pi} \frac{\sin 2nx}{p^2 - 2p \cos x + 1}$$

$$= \int_{-\pi}^{\pi} \frac{e^{2nx} \, dx}{p^2 - 2p \cos x + 1} = \int_{|z|=1} \frac{z^{2n}}{(p^2+1) - 2p \frac{z^2+1}{2z}} \frac{dz}{jz}$$

$$= -j \int_{|z|=1} \frac{z^{2n} \, dz}{-pz^2 + (p^2+1)z - p} = j \int_{|z|=1} \frac{z^{2n} \, dz}{pz^2 - (p^2+1)z + p}$$

$$= j 2\pi R_{20}$$

$$|p| > 1 \Rightarrow \left| \frac{1}{p} \right| < 1$$

$$R_{20} = R_{z_2} = \frac{z_2^{2n}}{2pz_2 - (p^2+1)} = \frac{\left(\frac{1}{p}\right)^{2n}}{2 - p^2 - 1} = -\frac{1}{p^{2n}(p^2-1)}$$

$$\text{Case I} \\ |p| > 1 \Rightarrow J = +2\pi \frac{1}{p^{2n}(p^2-1)} \in \mathbb{R}$$

$$\Rightarrow \int_{-\pi}^{\pi} \frac{\cos 2nx}{p^2 - 2p \cos x + 1} dx = \frac{2\pi}{p^{2n}(p^2 - 1)}$$

Caș II

$$|p| < 1$$

$$\Rightarrow z_0 = z_1$$

$$\int_{-\pi}^{\pi} \frac{dx}{p^2 - 2p \cos x + 1} = \oint \cdot 2\pi j \cdot \frac{1}{2p \cdot \frac{1}{p} - (p^2 + 1)}$$

$$\Rightarrow \int_{-\pi}^{\pi} \frac{\cos 2nx}{p^2 - 2p \cos x + 1} dx = \frac{2\pi}{p^2 - 1} - \frac{2\pi}{p^{2n}(p^2 - 1)} = \frac{2\pi}{(p^2 - 1)} \left(1 - \frac{1}{p^{2n}} \right)$$

$$3.6 \int_{-\infty}^{\infty} \frac{x^2 + 5x + 2}{(x^2 + 1)(x^2 + 4)} dx = 2\pi j \sum_{\substack{\text{Im } z_i > 0 \\ z_i - \text{poli simpli}}} \text{Res } z_i + \pi j \sum_{\substack{\text{Im } z_i = 0 \\ z_i - \text{poli simpli}}} \text{Res } z_i$$

\$z_i\$ - singularități din semiplanul superior.

- dacă poli de pe axa reală nu sunt simpli, integrala e 0

$$(z^2 + 1)(z^2 + 4)^2 \neq 0$$

$$z^2 + 1 = 0 \Rightarrow z = \pm j - \text{poli simpli}$$

$$(z^2 + 4)^2 = 0 \Rightarrow z = \pm 2j - \text{poli dubli}$$

$$I = 2\pi j (R_j + R_{2j})$$

$$R_j = \frac{z^2 + 5z + 2}{\underbrace{2z(z^2 + 4)^2 + (z^2 + 1)2(z^2 + 4) \cdot 2z}_{=0}} \Big|_{z=j}$$

$$= \frac{-1 + 5j + 2}{2j(-1 + 4)^2} = \frac{1 + 5j}{2j \cdot 9}$$

data z_0 - pol de ordin $k \Rightarrow \text{Res}(f; z_0) = \frac{1}{(k-1)!} \lim_{z \rightarrow z_0} (z - z_0)^{k-1} f(z)$

$$\text{Res}(f; 2j) = \lim_{z \rightarrow 2j} \cancel{(z - 2j)^2} \cdot \frac{z^2 + 5z + 2}{(z+1)\cancel{(z - 2j)^2}(z+2j)^2}$$

$$= \lim_{z \rightarrow 2j} \frac{(z+5)(z+1)(z+2j)^2 - (z^2 + 5z + 2)[2z(z+2j)^2 + 2(z+1)(z+2j)]}{(z+1)(z+2j)^4}$$

$$= \frac{(4j+5)(-4+1)(4j) - [(6+10j+2)(4j-4j+2(-4+1))]}{(-4+1)^2(4j)^2}$$

$$= \frac{-12j(4j+5) - (2+10j)(-16-6)}{-9 \cdot 16 \cdot 4j}$$

$$= \frac{48 - 60j + 44 + 220j}{-9 \cdot 16 \cdot 4j}$$

$$= \frac{92 + 160j}{-9 \cdot 16 \cdot 4j}$$

$$= \frac{23 + 40j}{-9 \cdot 16j} = -\frac{40 + 23j}{9 \cdot 16}$$

$$i = 2\pi j \left(\frac{1+5j}{32j} - \frac{1+40j}{9 \cdot 16 \cdot 9} \right)$$

$$= \frac{\pi}{9} \left(1+5j - \frac{1+40j}{8} \right)$$

$$= \frac{\pi}{72} (8 + \cancel{40j} - 1 - \cancel{40j}) = \frac{7\pi}{72} \in \mathbb{R} !$$

$$4^{\circ} I = \int_0^{\infty} \frac{x^2}{x^4 + x^2 + 1} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{y^2}{y^4 + y^2 + 1} dy$$

$$= \pi \int (R_{z_1} + R_{z_2})$$

$$z^4 + z^2 + 1 = 0$$

$$z^6 - 1 = (z^2 - 1)(z^4 + z^2 + 1)$$

$$\Rightarrow z^4 + z^2 + 1 = \frac{z^6 - 1}{z^2 - 1}$$

$$\Rightarrow \frac{z^6 - 1}{z^2 - 1} = 0$$

$$z^6 - 1 = 0$$

$$z^6 = e^{2k\pi j} = 1$$

$$\Rightarrow z_k = e^{\frac{2k\pi j}{6}} ; k = 0, 1, 2, 3, 4, 5$$

poli simple

$$\left\{ \begin{array}{l} z_0 = 1 \\ z_1 = \cos \frac{\pi}{3} + j \sin \frac{\pi}{3} = \frac{1}{2} + j \frac{\sqrt{3}}{2} \\ z_2 = e^{\frac{2\pi j}{3}} = \cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} = -\frac{1}{2} + j \frac{\sqrt{3}}{2} \\ z_3 = e^{\frac{\pi j}{2}} = \cos \frac{\pi}{2} + j \sin \frac{\pi}{2} = j \\ z_4 = e^{\frac{4\pi j}{3}} = \cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} = -\frac{1}{2} - j \frac{\sqrt{3}}{2} \\ z_5 = e^{\frac{5\pi j}{3}} = \cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3} = \frac{1}{2} - j \frac{\sqrt{3}}{2} \end{array} \right.$$

poli de ordin 1

↳ vârfurile hexagonului inscris în cerc

$$R_{z_1} = \frac{(z_1)^2}{4z_1^3 + 2z_1} = \frac{z_1^2}{4z_1^2 + 2}$$